1 Thermodynamic relaxation

1.1 Relaxation as entropy maximization

Per unit mass of a substance, the first law of thermodynamics states that

$$de = T\,ds - p\,dv. \tag{1}$$

Specific enthalpy and specific Gibbs energy are defined by

$$h \equiv e + pv, \qquad g \equiv h - Ts. \tag{2}$$

Differentiating the definitions with the product rule and using (1),

$$dh = de + p\,dv + v\,dp = T\,ds + v\,dp, \tag{3}$$

$$dg = dh - T\,ds - s\,dT = v\,dp - s\,dT. \tag{4}$$

With the extensive variables $E = me$, $S = ms$, $V = mv$,

$$dE = m\,de + e\,dm, \qquad dS = m\,ds + s\,dm, \qquad dV = m\,dv + v\,dm. \tag{5}$$

Therefore

$$\begin{aligned} T\,dS - p\,dV &= m\,(T\,ds - p\,dv) + (Ts - pv)\,dm = m\,de + (Ts - pv)\,dm \\ &= dE - (e + pv - Ts)\,dm = dE - g\,dm, \end{aligned} \tag{6}$$

that is,

$$dE = T\,dS - p\,dV + g\,dm. \tag{7}$$

Note that the above equation holds for single phase single species. For a multi-species mixture of single phase, where the species are at thermo-mechanical equilibrium we have

$$dE = T\,dS - p\,dV + \sum_{i=1}^{n} \mu_i\,dN_i, \tag{8}$$

where $\mu_i = \frac{\partial E}{\partial N_i}$ is defined as the chemical potential of i_{th} component in the mixture (the energy required to add one mole of species i to the mixture). Differentiating $G = E + pV - TS$ and using equation (8) we find that

$$\frac{\partial G}{\partial N_i} = \mu_i. \tag{9}$$

Assuming that in the mixture, species are molecularly mixed, the entropy of the mixture is modeled as

$$S = \sum_{i=1}^{n} m_i s_i - R \sum_{i=1}^{n} N_i \ln x_i, \tag{10}$$

where the first term is individual species entropy and the second term models the entropy of mixing. Correspondingly, the mixture Gibbs energy would be

$$G = \sum_{i=1}^{n} m_i g_i + RT \sum_{i=1}^{n} N_i \ln x_i, \tag{11}$$

where g_i is the gibbs energy of the component i per unit mass. Therefore we have

$$\left(\frac{\partial G}{\partial m_i}\right)_{p,T,m_{j\neq i}} = g_i + R_i T \ln x_i, \tag{12}$$

which allows us to re-write equation (8) as

$$dE = T\,dS - p\,dV + \sum_{i=1}^{n} \left(g_i + R_i T \ln x_i\right)\, dm_i, \tag{13}$$

which can be rearranged for S as

$$dS = \frac{p}{T}\,dV + \frac{1}{T}\,dE - \sum_{i=1}^{n} \left(\frac{g_i(p,T)}{T} + R_i \ln x_i\right)\, dm_i. \tag{14}$$

Defining the mass specific chemical potential $\hat{\mu}_i = \frac{\mu_i}{M_i}$, the above equation takes the following compact form

$$dS = \frac{p}{T}\,dV + \frac{1}{T}\,dE - \sum_{i=1}^{n} \frac{\hat{\mu}_i(p,T)}{T}\, dm_i, \tag{15}$$

where $\hat{\mu}_i = g_i + R_i T \ln x_i$. For the single species liquid phase we have

$$dS_l = \frac{p_l}{T_l}\,dV_l + \frac{1}{T_l}\,dE_l - \frac{g_l(p_l,T_l)}{T_l}\, dm_l. \tag{16}$$

For the gas mixture, using ideal gas EOS, we can simplify the entropy differential relation. For pure vapor, at any given p and T, equation (4) reads

$$\left(\frac{\partial g_v}{\partial p}\right)_T = v_v = \frac{R_v T}{p}, \tag{17}$$

where the ideal gas EOS has been used to express v_v in terms of p, T, and R_v. Integrating the above equation at constant T gives

$$\int_{p_g}^{p_v} \left(\frac{\partial g_v}{\partial p}\right)_T dp = \int_{p_g}^{p_v} \frac{R_v T}{p}\, dp = R_v T \ln\left(\frac{p_v}{p_g}\right), \tag{18}$$

which leads to the following relation

$$g_v\left(p_v, T_g\right) - g_v\left(p_g, T_g\right) = R_v T \ln x_v. \tag{19}$$

The above relation can be plugged into equation (14) for the ideal gas mixture to get a simplified relation for dS_g

$$dS_g = \frac{p_g}{T_g}\,dV_g + \frac{1}{T_g}\,dE_g - \frac{g_v(p_v,T_g)}{T_g}\, dm_v. \tag{20}$$

The extensive thermodynamic state for phase k is defined as

$$\boldsymbol{x_k} = [V_k,\ E_k,\ m_k]^T, \tag{21}$$

where in general, m_k is a vector of transferable species masses. Here, since liquid is assumed to be single species and gas is a mixture of vapor and air only, m_k would be a scalar quantity taking m_l

for the liquid and m_v for the gas phase. Corresponding to the state vector for phase k, potentials are the partial derivatives of the entropy,

$$\boldsymbol{y_k} \equiv \frac{\partial S_k}{\partial \boldsymbol{x_k}} = \left[\frac{p_k}{T_k},\ \frac{1}{T_k},\ -\frac{\hat{\mu_k}}{T_k}\right]^T, \tag{22}$$

where the same argument for m_k applies to μ_k.
Relaxation to thermodynamic equilibrium is a process in which, due to the potential ($\boldsymbol{y}$) mismatch between liquid and gas phases, the extensive state ($\boldsymbol{x}$) is exchanged between the two phases until the potentials equilibrate. Volume (together with the work it carries), heat, and mass are transferred during mechanical, thermal, and chemical relaxation processes, respectively. During a cell local relaxation process, the total extensive thermodynamic quantities $\boldsymbol{x}$ are conserved via

$$\sum_{k=1}^{n_p} d\boldsymbol{x}_k = 0. \tag{23}$$

For a two-phase system of liquid and gas, define $\boldsymbol{\xi} \equiv \delta \boldsymbol{x}_l$ being the change in x_l that takes the initial state to equilibrium state and therefore, $\delta \boldsymbol{x}_g = -\boldsymbol{\xi}$. Plugging equations (21) and (22) into equation (14), the entropy gain corresponding to relaxation would be

$$dS = \boldsymbol{F} \cdot d\boldsymbol{x}_l, \tag{24}$$

where $\boldsymbol{F} \equiv \boldsymbol{y}_l - \boldsymbol{y}_g$ is the thermodynamic driving force. Instantaneous relaxation to equilibrium jumps entropy to its maximum value over the exchanges allowed by (23). Therefore, the maximum is where the derivative of total entropy with respect to its free parameter is zero. We choose $\boldsymbol{x}_l$ to be the free parameter, over which we are maximizing S. Differentiating S with respect to $\boldsymbol{x}_l$ we get

$$\frac{\partial S}{\partial \boldsymbol{x}_l} = \frac{\partial S_l}{\partial \boldsymbol{x}_l} + \frac{\partial S_g}{\partial \boldsymbol{x}_g}\frac{\partial \boldsymbol{x}_g}{\partial \boldsymbol{x}_l} = \boldsymbol{y}_l - \boldsymbol{y}_g = \boldsymbol{F} \tag{25}$$

where equations (22) and (23) are used to apply chain rule. The above relation restates the equilibrium condition being equivalent to equilibration of phase potentials or zero remaining thermodynamic force. Note that if S is strictly concave, $\boldsymbol{\xi} = \boldsymbol{x}_l^{eq} - \boldsymbol{x}_l^0$ is the only solution that maximizes S.

1.1.1 Surface tension

Let the interface have area A and tension σ, independent of temperature, so that it carries energy σA and no entropy. When the liquid volume changes, $dA = \kappa_c\, dV_l$, where κ_c is the interface curvature. The conserved energy is therefore $E_l + E_g + \sigma A$, so

$$dE_g = -dE_l - \sigma\kappa_c\, dV_l, \qquad dV_g = -dV_l, \qquad dm_v = -dm_l. \tag{26}$$

Substituting into $dS = \boldsymbol{y}_l \cdot d\boldsymbol{x}_l + \boldsymbol{y}_g \cdot d\boldsymbol{x}_g$,

$$dS = \Big(\frac{p_l}{T_l} - \frac{p_g}{T_g} - \frac{p_j}{T_g}\Big)dV_l + \Big(\frac{1}{T_l} - \frac{1}{T_g}\Big)dE_l - \Big(\frac{g_l}{T_l} - \frac{g_v}{T_g}\Big)dm_l, \tag{27}$$

where the interface pressure jump is $p_j = \sigma\kappa_c$. In matrix form the constraints read $d\boldsymbol{x}_g = -\boldsymbol{A}\, d\boldsymbol{x}_l$ with

$$\boldsymbol{A} = \begin{bmatrix} 1 & 0 & 0 \\ p_j & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \tag{28}$$

so that $dS = \boldsymbol{y}_l \cdot d\boldsymbol{x}_l - \boldsymbol{y}_g \cdot \boldsymbol{A}\, d\boldsymbol{x}_l = (\boldsymbol{y}_l - \boldsymbol{A}^T \boldsymbol{y}_g) \cdot d\boldsymbol{x}_l$, which is (27). With surface tension the force in (24) is therefore

$$\boldsymbol{F} = \boldsymbol{y}_l - \boldsymbol{A}^T \boldsymbol{y}_g, \tag{29}$$

which reduces to $\boldsymbol{y}_l - \boldsymbol{y}_g$ for $p_j = 0$; below, $\boldsymbol{F}$ denotes (29). Equilibrium then requires

$$p_l - p_g = p_j, \qquad T_l = T_g, \qquad g_l(p_l, T) = g_v(p_v, T). \tag{30}$$

The first condition is the Laplace law. The third one keeps its form, but the liquid chemical potential is evaluated at the raised liquid pressure (Kelvin effect).

1.1.2 Solution to equilibrium

Define stiffness matrix for phase k as the Hessian of $-S_k$

$$\boldsymbol{K}_k = -\frac{\partial \boldsymbol{y}_k}{\partial \boldsymbol{x}_k}, \tag{31}$$

which is symmetric by definition and positive-semi definite given S_k being concave. Note that the entities of stiffness matrix are the reciprocal capacities of various types. Taylor series expansion of y_k:

$$\delta \boldsymbol{y}_k^n = \left.\frac{\partial \boldsymbol{y}_k}{\partial \boldsymbol{x}_k}\right|_{\boldsymbol{x}_k = \boldsymbol{x}_k^n} \delta \boldsymbol{x}_k^n + \mathcal{O}\left(\|\delta \boldsymbol{x}_k^n\|^2\right), \tag{32}$$

where $\delta \boldsymbol{y}_k^n = \boldsymbol{y}_k^{n+1} - \boldsymbol{y}_k^n$ and $\delta \boldsymbol{x}_k^n = \boldsymbol{x}_k^{n+1} - \boldsymbol{x}_k^n$. Subtracting $\boldsymbol{A}^T$ times the above equation written for gas from that written for liquid, and using the conservation constraint $\delta \boldsymbol{x}_g^n = -\boldsymbol{A}\, \delta \boldsymbol{x}_l^n$ with $\boldsymbol{A}$ constant,

$$\boldsymbol{F}^{n+1} - \boldsymbol{F}^n = -\left.\left(\boldsymbol{K}_l + \boldsymbol{A}^T \boldsymbol{K}_g \boldsymbol{A}\right)\right|_{\boldsymbol{x}_k = \boldsymbol{x}_k^n} \cdot \delta \boldsymbol{x}_l^n + \mathcal{O}\left(\|\delta \boldsymbol{x}_k^n\|^2\right), \tag{33}$$

Equilibrium requires $\boldsymbol{F}^{n+1} = 0$. Therefore, solution to equilibrium is obtained from iteratively solving the following equation

$$\boldsymbol{J}^n \cdot \delta \boldsymbol{x}_l^n = -\boldsymbol{F}^n, \tag{34}$$

where Jacobian of the n_{th} iteration is $\boldsymbol{J}^n = -\boldsymbol{K}^n$, with $\boldsymbol{K} \equiv \boldsymbol{K}_l + \boldsymbol{A}^T \boldsymbol{K}_g \boldsymbol{A}$ evaluated at $\boldsymbol{x}_k = \boldsymbol{x}_k^n$; $\boldsymbol{K} = \boldsymbol{K}_l + \boldsymbol{K}_g$ for $p_j = 0$. At each iteration, the intermediate solution is

$$\boldsymbol{x}_l^{n+1} = \boldsymbol{x}_l^n + \delta \boldsymbol{x}_l^n, \qquad \boldsymbol{x}_g^{n+1} = \boldsymbol{x}_g^n - \boldsymbol{A}\, \delta \boldsymbol{x}_l^n, \tag{35}$$

$$\delta \boldsymbol{y}_l^n = -\boldsymbol{\Phi}_l^n \cdot \boldsymbol{F}^n, \quad \boldsymbol{A}^T \delta \boldsymbol{y}_g^n = +\boldsymbol{\Phi}_g^n \cdot \boldsymbol{F}^n, \tag{36}$$

$$\boldsymbol{\Phi}_l^n \equiv \boldsymbol{K}_l^n (\boldsymbol{K}^n)^{-1}, \qquad \boldsymbol{\Phi}_g^n \equiv \boldsymbol{A}^T \boldsymbol{K}_g^n \boldsymbol{A} (\boldsymbol{K}^n)^{-1}; \qquad \boldsymbol{\Phi}_l^n + \boldsymbol{\Phi}_g^n = \boldsymbol{I}. \tag{37}$$

Definiteness of the stiffness matrices. The entropy of the liquid is extensive, $S_l(\lambda \boldsymbol{x}_l) = \lambda S_l(\boldsymbol{x}_l)$ for $\lambda > 0$. Differentiating with respect to $\boldsymbol{x}_l$ gives $\boldsymbol{y}_l(\lambda \boldsymbol{x}_l) = \boldsymbol{y}_l(\boldsymbol{x}_l)$, and differentiating this with respect to λ at $\lambda = 1$ gives $\boldsymbol{K}_l \boldsymbol{x}_l = 0$: $\boldsymbol{K}_l$ is singular, with null vector $\boldsymbol{x}_l$. For a stable equation of state ($c_v > 0$, $K_T > 0$) the specific entropy $s_l(v_l, e_l)$ is strictly concave, and $\boldsymbol{x}_l$ is the only null direction, so $\boldsymbol{K}_l$ is positive semi-definite of rank two. The gas entropy is extensive in $[V_g, E_g, m_v, m_a]$, and its 4×4 Hessian has the single null vector $[V_g, E_g, m_v, m_a]^T$. $\boldsymbol{K}_g$ is the leading 3×3 block of this Hessian, since the air is not exchanged. If $m_a > 0$ it is positive definite: a vector $\boldsymbol{z} \neq 0$ with $\boldsymbol{z}^T \boldsymbol{K}_g \boldsymbol{z} = 0$, extended by a zero m_a entry, would be a null direction of the full Hessian that is not parallel to $[V_g, E_g, m_v, m_a]^T$. Since $\boldsymbol{A}$ is invertible, for $\boldsymbol{z} \neq 0$

$$\boldsymbol{z}^T \boldsymbol{K} \boldsymbol{z} = \boldsymbol{z}^T \boldsymbol{K}_l \boldsymbol{z} + (\boldsymbol{A}\boldsymbol{z})^T \boldsymbol{K}_g (\boldsymbol{A}\boldsymbol{z}) > 0, \tag{38}$$

so $\boldsymbol{K}$ is positive definite at every state: (34) has a unique solution at every iterate, S is strictly concave in $\boldsymbol{x}_l$, and $\boldsymbol{F} = 0$ has at most one solution. For a gas of pure vapor ($m_a = 0$) $\boldsymbol{K}_g$ is singular with null vector $\boldsymbol{x}_g$, and $\boldsymbol{K}$ remains positive definite unless $\boldsymbol{x}_l$ is parallel to $\boldsymbol{A}^{-1}\boldsymbol{x}_g$, i.e. unless the two phases have the same specific state.

1.1.3 Matrix decomposition

Define the direction vectors:

$$\boldsymbol{d}_M = [1,\ -p_I,\ 0]^T \quad , \quad \delta\boldsymbol{x}_l^M = \delta\tilde{V}_l\,\boldsymbol{d}_M \tag{39}$$

$$\boldsymbol{d}_T = [0,\ -1,\ 0]^T \quad , \quad \delta\boldsymbol{x}_l^T = \delta\tilde{E}_l\,\boldsymbol{d}_T \tag{40}$$

$$\boldsymbol{d}_C = [-v_l,\ -e_l,\ -1]^T \quad , \quad \delta\boldsymbol{x}_l^C = \delta\tilde{m}_l\,\boldsymbol{d}_C \tag{41}$$

$$\delta\boldsymbol{x}_l = \delta\boldsymbol{x}_l^M + \delta\boldsymbol{x}_l^T + \delta\boldsymbol{x}_l^C \tag{42}$$

$$\delta V_l = \delta\tilde{V}_l - v_l\,\delta\tilde{m}_l, \quad \delta E_l = -p_I\,\delta\tilde{V}_l - \delta\tilde{E}_l - e_l\,\delta\tilde{m}_l, \quad \delta m_l = -\delta\tilde{m}_l. \tag{43}$$

$$\boldsymbol{D} = [\boldsymbol{d}_M,\ \boldsymbol{d}_T,\ \boldsymbol{d}_C] \tag{44}$$

$$\delta\boldsymbol{x}_l = \boldsymbol{D}\,\delta\tilde{\boldsymbol{x}}_l, \quad \delta\tilde{\boldsymbol{x}}_l = \left[\delta\tilde{V}_l,\ \delta\tilde{E}_l,\ \delta\tilde{m}_l\right]^T \tag{45}$$

$$\boldsymbol{J}^n \cdot \boldsymbol{D}^n\,\delta\tilde{\boldsymbol{x}}_l^n = -\boldsymbol{F}^n, \tag{46}$$

Choice of directions. The three columns of $\boldsymbol{D}$ are the three transfers: volume received by the liquid with its work at p_I, heat given by the liquid, and mass given by the liquid. Each transfer is counted positive in the direction in which its liquid-minus-gas force drives it: the liquid expands when its pressure is higher, and gives heat and mass when its temperature and chemical potential are higher. $\det \boldsymbol{D} = 1$, so every $\delta\boldsymbol{x}_l$ has a unique decomposition. In the sequential relaxation below, each step holds the equilibria of the faster ones, so the steps move within the nested sets $\mathrm{span}\{\boldsymbol{d}_M\} \subset \mathrm{span}\{\boldsymbol{d}_M, \boldsymbol{d}_T\} \subset \mathbb{R}^3$, not along $\boldsymbol{d}_T$ and $\boldsymbol{d}_C$ themselves. $\mathrm{span}\{\boldsymbol{d}_M, \boldsymbol{d}_T\} = \{\delta m_l = 0\}$ does not depend on the choice of $\boldsymbol{d}_T$. Replacing $\boldsymbol{d}_C$ by $\boldsymbol{d}_C + a\,\boldsymbol{d}_M + b\,\boldsymbol{d}_T$ does not change the chemical step either, because its δV_l and δE_l follow from the held equilibria. The one choice that changes a sequential result is p_I in $\boldsymbol{d}_M$, which sets the mechanical end state.

Newton step in the transfer coordinates. The directions are evaluated at a reference state $\boldsymbol{x}^r$, $\boldsymbol{D} = \boldsymbol{D}(\boldsymbol{x}^r)$, through p_I, v_l and e_l. The transfer coordinates $\tilde{\boldsymbol{x}}_l = [\tilde{V}_l, \tilde{E}_l, \tilde{m}_l]^T$ are defined by the linear change of variables

$$\boldsymbol{x}_l = \boldsymbol{x}_l^r + \boldsymbol{D}\,\tilde{\boldsymbol{x}}_l, \tag{47}$$

with $\boldsymbol{D}$ held constant, so that $\tilde{\boldsymbol{x}}_l = 0$ at $\boldsymbol{x}^r$. $\tilde{V}_l$ is the volume (with its work) received by the liquid, and $\tilde{E}_l$ and $\tilde{m}_l$ are the heat and the mass given by the liquid to the gas. Differentiating (47),

$$d\boldsymbol{x}_l = \boldsymbol{D}\,d\tilde{\boldsymbol{x}}_l. \tag{48}$$

Entropy. Substituting (48) into (24) and exchanging the order of summation,

$$dS = \sum_i F_i\,dx_{l,i} = \sum_i F_i \sum_j D_{ij}\,d\tilde{x}_{l,j} = \sum_j \Big(\sum_i D_{ij}F_i\Big) d\tilde{x}_{l,j} = \left(\boldsymbol{D}^T\boldsymbol{F}\right) \cdot d\tilde{\boldsymbol{x}}_l = \tilde{\boldsymbol{F}} \cdot d\tilde{\boldsymbol{x}}_l, \tag{49}$$

where $\tilde{\boldsymbol{F}} \equiv \boldsymbol{D}^T\boldsymbol{F} = \partial S/\partial\tilde{\boldsymbol{x}}_l$. $\tilde{\boldsymbol{F}} = [\tilde{F}_M, \tilde{F}_T, \tilde{F}_C]^T$ is therefore the entropy gained per unit of volume, heat and mass transferred: the forces conjugate to the three transfers.

Force. By (31), $d\boldsymbol{y}_l = -\boldsymbol{K}_l\, d\boldsymbol{x}_l$ and $d\boldsymbol{y}_g = -\boldsymbol{K}_g\, d\boldsymbol{x}_g = \boldsymbol{K}_g\boldsymbol{A}\, d\boldsymbol{x}_l$, since $d\boldsymbol{x}_g = -\boldsymbol{A}\, d\boldsymbol{x}_l$ with $\boldsymbol{A}$ constant. Differentiating $\boldsymbol{F} = \boldsymbol{y}_l - \boldsymbol{A}^T\boldsymbol{y}_g$,

$$d\boldsymbol{F} = d\boldsymbol{y}_l - \boldsymbol{A}^T d\boldsymbol{y}_g = -(\boldsymbol{K}_l + \boldsymbol{A}^T\boldsymbol{K}_g\boldsymbol{A})\, d\boldsymbol{x}_l = -\boldsymbol{K}\, d\boldsymbol{x}_l. \tag{50}$$

Multiplying (50) by the constant $\boldsymbol{D}^T$ and substituting (48),

$$d\tilde{\boldsymbol{F}} = \boldsymbol{D}^T d\boldsymbol{F} = -\boldsymbol{D}^T\boldsymbol{K}\boldsymbol{D}\, d\tilde{\boldsymbol{x}}_l = -\tilde{\boldsymbol{K}}\, d\tilde{\boldsymbol{x}}_l, \qquad \tilde{\boldsymbol{K}} \equiv \boldsymbol{D}^T\boldsymbol{K}\boldsymbol{D}. \tag{51}$$

$\tilde{\boldsymbol{K}} = -\partial\tilde{\boldsymbol{F}}/\partial\tilde{\boldsymbol{x}}_l = -\partial^2 S/\partial\tilde{\boldsymbol{x}}_l^2$ is the stiffness matrix in the transfer coordinates. It is symmetric, $\tilde{\boldsymbol{K}}^T = \boldsymbol{D}^T\boldsymbol{K}^T\boldsymbol{D} = \tilde{\boldsymbol{K}}$, because $\boldsymbol{K}$ is. It is positive definite when $\boldsymbol{K}$ is: for $\boldsymbol{z} \neq 0$, $\boldsymbol{D}\boldsymbol{z} \neq 0$ and $\boldsymbol{z}^T\tilde{\boldsymbol{K}}\boldsymbol{z} = (\boldsymbol{D}\boldsymbol{z})^T\boldsymbol{K}(\boldsymbol{D}\boldsymbol{z}) > 0$.

Newton–Raphson. Let $\delta\tilde{\boldsymbol{x}}_l^n = [\delta\tilde{V}_l^n, \delta\tilde{E}_l^n, \delta\tilde{m}_l^n]^T$ be the transfer of iteration n, which takes $\boldsymbol{x}^n$ to $\boldsymbol{x}^{n+1}$. Taking the reference state at the current iterate, $\boldsymbol{x}^r = \boldsymbol{x}^n$ and $\boldsymbol{D}^n \equiv \boldsymbol{D}(\boldsymbol{x}^n)$, and Taylor expanding $\tilde{\boldsymbol{F}}$ about $\boldsymbol{x}^n$ with (51),

$$\tilde{\boldsymbol{F}}^{n+1} = \tilde{\boldsymbol{F}}^n - \tilde{\boldsymbol{K}}^n\, \delta\tilde{\boldsymbol{x}}_l^n + \mathcal{O}\big(\|\delta\tilde{\boldsymbol{x}}_l^n\|^2\big), \tag{52}$$

with $\tilde{\boldsymbol{F}}^n = (\boldsymbol{D}^n)^T\boldsymbol{F}^n$, $\tilde{\boldsymbol{F}}^{n+1} = (\boldsymbol{D}^n)^T\boldsymbol{F}^{n+1}$, and $\tilde{\boldsymbol{K}}^n$ evaluated at $\boldsymbol{x}^n$. $\boldsymbol{D}^n$ is invertible ($\det\boldsymbol{D}^n = 1$), so $\boldsymbol{F}^{n+1} = 0$ if and only if $\tilde{\boldsymbol{F}}^{n+1} = 0$. Setting $\tilde{\boldsymbol{F}}^{n+1} = 0$ in (52) and dropping the second-order term, the solution to equilibrium is obtained from iteratively solving

$$\tilde{\boldsymbol{J}}^n \cdot \delta\tilde{\boldsymbol{x}}_l^n = -\tilde{\boldsymbol{F}}^n, \qquad \tilde{\boldsymbol{J}}^n \equiv -\tilde{\boldsymbol{K}}^n = (\boldsymbol{D}^n)^T\boldsymbol{J}^n\boldsymbol{D}^n, \tag{53}$$

and at each iteration the intermediate solution is

$$\boldsymbol{x}_l^{n+1} = \boldsymbol{x}_l^n + \boldsymbol{D}^n\, \delta\tilde{\boldsymbol{x}}_l^n, \qquad \boldsymbol{x}_g^{n+1} = \boldsymbol{x}_g^n - \boldsymbol{A}\boldsymbol{D}^n\, \delta\tilde{\boldsymbol{x}}_l^n. \tag{54}$$

(53) is (46) written in the transfer coordinates.

Shorter route. The same system follows directly from (33). Setting $\boldsymbol{F}^{n+1} = 0$ in (33) and dropping the second-order term,

$$-\boldsymbol{F}^n = -\boldsymbol{K}^n\, \delta\boldsymbol{x}_l^n. \tag{55}$$

Substituting $\delta\boldsymbol{x}_l^n = \boldsymbol{D}^n\, \delta\tilde{\boldsymbol{x}}_l^n$ and multiplying from the left by $(\boldsymbol{D}^n)^T$,

$$-(\boldsymbol{D}^n)^T\boldsymbol{F}^n = -(\boldsymbol{D}^n)^T\boldsymbol{K}^n\boldsymbol{D}^n\, \delta\tilde{\boldsymbol{x}}_l^n \quad \Longrightarrow \quad \tilde{\boldsymbol{J}}^n \cdot \delta\tilde{\boldsymbol{x}}_l^n = -\tilde{\boldsymbol{F}}^n, \tag{56}$$

by the definitions of $\tilde{\boldsymbol{F}}$ in (49) and of $\tilde{\boldsymbol{K}}$ in (51). This is (53). The derivation from the differentials above shows in addition that $\tilde{\boldsymbol{F}} = \partial S/\partial\tilde{\boldsymbol{x}}_l$ and $\tilde{\boldsymbol{K}} = -\partial^2 S/\partial\tilde{\boldsymbol{x}}_l^2$, which the shorter route does not.

In the Newton systems below, the superscript n marks quantities evaluated at the current iterate $\boldsymbol{x}^n$; the differential relations hold at any state and carry no superscript.

Derivation of the conjugate forces. The components of $\boldsymbol{F}$ are the coefficients of dV_l, dE_l and dm_l in (27),

$$F_1 = \frac{p_l}{T_l} - \frac{p_g + p_j}{T_g}, \qquad F_2 = \frac{1}{T_l} - \frac{1}{T_g}, \qquad F_3 = -\frac{g_l}{T_l} + \frac{g_v}{T_g}, \tag{57}$$

with $g_l = g_l(p_l, T_l)$ and $g_v = g_v(p_v, T_g)$. The rows of $\boldsymbol{D}^T$ are $\boldsymbol{d}_M^T$, $\boldsymbol{d}_T^T$, $\boldsymbol{d}_C^T$,

$$\boldsymbol{D}^T = \begin{bmatrix} 1 & -p_I & 0 \\ 0 & -1 & 0 \\ -v_l & -e_l & -1 \end{bmatrix}, \tag{58}$$

where $v_l = V_l/m_l$ and $e_l = E_l/m_l$ are the specific volume and specific internal energy of the liquid. The product $\tilde{\boldsymbol{F}} = \boldsymbol{D}^T\boldsymbol{F}$ is therefore

$$\tilde{F}_M = F_1 - p_I F_2, \qquad \tilde{F}_T = -F_2, \qquad \tilde{F}_C = -(v_l F_1 + e_l F_2 + F_3). \tag{59}$$

Mechanical. Substituting F_1 and F_2 from (57),

$$\tilde{F}_M = \frac{p_l}{T_l} - \frac{p_g + p_j}{T_g} - \frac{p_I}{T_l} + \frac{p_I}{T_g}, \tag{60}$$

and grouping the terms over T_l and over T_g,

$$\tilde{F}_M = \frac{p_l - p_I}{T_l} + \frac{p_I - p_g - p_j}{T_g}. \tag{61}$$

Thermal. $\tilde{F}_T = -F_2 = 1/T_g - 1/T_l$ directly.

Chemical. Substituting F_1, F_2 and F_3 from (57),

$$\tilde{F}_C = -v_l\Big(\frac{p_l}{T_l} - \frac{p_g + p_j}{T_g}\Big) - e_l\Big(\frac{1}{T_l} - \frac{1}{T_g}\Big) + \frac{g_l}{T_l} - \frac{g_v}{T_g}, \tag{62}$$

and grouping the terms over T_l and over T_g,

$$\tilde{F}_C = -\frac{e_l + p_l v_l - g_l}{T_l} + \frac{e_l + (p_g + p_j)v_l - g_v}{T_g}. \tag{63}$$

By (2), $h_l = e_l + p_l v_l$ and $g_l = h_l - T_l s_l$, so the numerator of the first term is

$$e_l + p_l v_l - g_l = h_l - g_l = T_l s_l, \tag{64}$$

and the first term of (63) is $-s_l$. The second term does not simplify the same way, because v_l is multiplied by $p_g + p_j$ rather than by p_l, and g_v is evaluated at T_g.

Collecting the three results,

$$\tilde{F}_M = \frac{p_l - p_I}{T_l} + \frac{p_I - p_g - p_j}{T_g}, \qquad \tilde{F}_T = \frac{1}{T_g} - \frac{1}{T_l}, \qquad \tilde{F}_C = \frac{e_l + (p_g + p_j)v_l - g_v}{T_g} - s_l. \tag{65}$$

Forces of the sequential steps. The literature, e.g. [1], writes the three steps in the differences of pressure, temperature and chemical potential,

$$\hat{F}_M \equiv p_l - p_g - p_j, \qquad \hat{F}_T \equiv T_l - T_g, \qquad \hat{F}_C \equiv g_l - g_v, \tag{66}$$

collected in $\hat{\boldsymbol{F}} \equiv [\hat{F}_M, \hat{F}_T, \hat{F}_C]^T$. With the intensive variables of each phase, $\boldsymbol{z}_l = [p_l, T_l, g_l]^T$ and $\boldsymbol{z}_g = [p_g, T_g, g_v]^T$, $g_v = g_v(p_v, T_g)$,

$$\hat{\boldsymbol{F}} = \boldsymbol{z}_l - \boldsymbol{z}_g - [p_j,\, 0,\, 0]^T. \tag{67}$$

$\hat{\boldsymbol{F}}$ is not $\tilde{\boldsymbol{F}}$: $\tilde{\boldsymbol{F}}$ holds derivatives of the entropy, $\hat{\boldsymbol{F}}$ differences of intensive variables. Below, the Newton iteration is derived with $\hat{\boldsymbol{F}}$ as the residual, following the steps of (50)–(53), and the two residuals are then related.

Newton step in $\hat{\boldsymbol{F}}$. *Intensive variables from the potentials.* By (22), the potentials of phase k are $\boldsymbol{y}_k = [p_k/T_k,\, 1/T_k,\, -g_k/T_k]^T$, with $g_g \equiv g_v$ as in (57). Inverting these relations,

$$p_k = \frac{y_{k,1}}{y_{k,2}}, \qquad T_k = \frac{1}{y_{k,2}}, \qquad g_k = -\frac{y_{k,3}}{y_{k,2}}. \tag{68}$$

Differentiating, and using $1/y_{k,2} = T_k$, $y_{k,1}/y_{k,2}^2 = p_k T_k$ and $y_{k,3}/y_{k,2}^2 = -g_k T_k$,

$$\begin{aligned} dp_k &= \frac{dy_{k,1}}{y_{k,2}} - \frac{y_{k,1}}{y_{k,2}^2}\, dy_{k,2} = T_k\, dy_{k,1} - p_k T_k\, dy_{k,2}, \\ dT_k &= -\frac{dy_{k,2}}{y_{k,2}^2} = -T_k^2\, dy_{k,2}, \\ dg_k &= -\frac{dy_{k,3}}{y_{k,2}} + \frac{y_{k,3}}{y_{k,2}^2}\, dy_{k,2} = -g_k T_k\, dy_{k,2} - T_k\, dy_{k,3}, \end{aligned} \tag{69}$$

or, in matrix form,

$$d\boldsymbol{z}_k = \boldsymbol{G}_k\, d\boldsymbol{y}_k, \qquad \boldsymbol{G}_k \equiv \frac{\partial \boldsymbol{z}_k}{\partial \boldsymbol{y}_k} = T_k \begin{bmatrix} 1 & -p_k & 0 \\ 0 & -T_k & 0 \\ 0 & -g_k & -1 \end{bmatrix}. \tag{70}$$

Force. As in (50), $d\boldsymbol{y}_l = -\boldsymbol{K}_l\, d\boldsymbol{x}_l$ and $d\boldsymbol{y}_g = \boldsymbol{K}_g \boldsymbol{A}\, d\boldsymbol{x}_l$, and p_j is constant, like $\boldsymbol{A}$. Differentiating (67) and substituting (70) and (48),

$$d\hat{\boldsymbol{F}} = \boldsymbol{G}_l\, d\boldsymbol{y}_l - \boldsymbol{G}_g\, d\boldsymbol{y}_g = -(\boldsymbol{G}_l \boldsymbol{K}_l + \boldsymbol{G}_g \boldsymbol{K}_g \boldsymbol{A})\, d\boldsymbol{x}_l = \hat{\boldsymbol{J}}\, d\tilde{\boldsymbol{x}}_l, \qquad \hat{\boldsymbol{J}} \equiv -(\boldsymbol{G}_l \boldsymbol{K}_l + \boldsymbol{G}_g \boldsymbol{K}_g \boldsymbol{A})\boldsymbol{D}. \tag{71}$$

$\hat{\boldsymbol{J}} = \partial \hat{\boldsymbol{F}}/\partial \tilde{\boldsymbol{x}}_l$ is the Jacobian of $\hat{\boldsymbol{F}}$ in the transfer coordinates. Unlike $\tilde{\boldsymbol{J}}$ it is not symmetric, because $\boldsymbol{G}_k \neq \boldsymbol{I}$.

Newton–Raphson. With $\boldsymbol{x}^r = \boldsymbol{x}^n$ as in (52), so that $\boldsymbol{D} = \boldsymbol{D}^n$ in $\hat{\boldsymbol{J}}^n$, and Taylor expanding $\hat{\boldsymbol{F}}$ about $\boldsymbol{x}^n$ with (71),

$$\hat{\boldsymbol{F}}^{n+1} = \hat{\boldsymbol{F}}^n + \hat{\boldsymbol{J}}^n\, \delta\tilde{\boldsymbol{x}}_l^n + \mathcal{O}\big(\|\delta\tilde{\boldsymbol{x}}_l^n\|^2\big). \tag{72}$$

By (77) below, $\hat{\boldsymbol{F}} = 0$ if and only if $\tilde{\boldsymbol{F}} = 0$, i.e. $\boldsymbol{F} = 0$. Setting $\hat{\boldsymbol{F}}^{n+1} = 0$ in (72) and dropping the second-order term, the solution to equilibrium is obtained from iteratively solving

$$\hat{\boldsymbol{J}}^n \cdot \delta\tilde{\boldsymbol{x}}_l^n = -\hat{\boldsymbol{F}}^n, \tag{73}$$

with the update (54). The unknowns are the transfers of (53); only the residual is different.

Relation between $\tilde{\boldsymbol{F}}$ and $\hat{\boldsymbol{F}}$. Each component of (65) is rewritten in $\hat{F}_M$, $\hat{F}_T$ and $\hat{F}_C$.

Thermal. Over the common denominator, $\tilde{F}_T = (T_l - T_g)/(T_l T_g) = \hat{F}_T/(T_l T_g)$.

Mechanical. Substituting $p_I - p_g - p_j = \hat{F}_M - (p_l - p_I)$ in (65),

$$\tilde{F}_M = \frac{p_l - p_I}{T_l} + \frac{\hat{F}_M - (p_l - p_I)}{T_g} = \frac{\hat{F}_M}{T_g} + (p_l - p_I)\Big(\frac{1}{T_l} - \frac{1}{T_g}\Big) = \frac{\hat{F}_M}{T_g} - \frac{p_l - p_I}{T_l T_g}\, \hat{F}_T. \tag{74}$$

Chemical. Substituting $p_g + p_j = p_l - \hat{F}_M$ and $g_v = g_l - \hat{F}_C$, and $e_l + p_l v_l - g_l = T_l s_l$ by (2), the numerator of the first term of $\tilde{F}_C$ in (65) is

$$e_l + (p_g + p_j)v_l - g_v = e_l + p_l v_l - g_l - v_l \hat{F}_M + \hat{F}_C = T_l s_l - v_l \hat{F}_M + \hat{F}_C, \tag{75}$$

so

$$\tilde{F}_C = \frac{T_l s_l - v_l \hat{F}_M + \hat{F}_C}{T_g} - s_l = s_l \frac{T_l - T_g}{T_g} + \frac{\hat{F}_C - v_l \hat{F}_M}{T_g} = -\frac{v_l}{T_g}\hat{F}_M + \frac{s_l}{T_g}\hat{F}_T + \frac{1}{T_g}\hat{F}_C. \tag{76}$$

Collecting the three results,

$$\tilde{\boldsymbol{F}} = \boldsymbol{C}\,\hat{\boldsymbol{F}}, \qquad \boldsymbol{C} = \frac{1}{T_g}\begin{bmatrix} 1 & -(p_l - p_I)/T_l & 0 \\ 0 & 1/T_l & 0 \\ -v_l & s_l & 1 \end{bmatrix}, \tag{77}$$

which holds at any state. $\det \boldsymbol{C} = 1/(T_l T_g^3) > 0$, so $\tilde{\boldsymbol{F}} = 0$ if and only if $\hat{\boldsymbol{F}} = 0$: (53) and (73) have the same root.

Forces of the steps. The interface pressure lies between $p_g + p_j$ and p_l; write it as

$$p_l - p_I = w_M \hat{F}_M, \qquad p_I - p_g - p_j = (1 - w_M)\hat{F}_M, \qquad w_M \in [0, 1]. \tag{78}$$

The first row of (77) is then $\tilde{F}_M = \hat{F}_M/T_g - w_M \hat{F}_M (T_l - T_g)/(T_l T_g) = \hat{F}_M\,[w_M/T_l + (1 - w_M)/T_g]$, and the third row at $\hat{F}_M = \hat{F}_T = 0$ with $T_l = T_g = T$ is $\hat{F}_C/T$:

$$\begin{gathered} \tilde{F}_M = \frac{\hat{F}_M}{T_M}, \quad \frac{1}{T_M} \equiv \frac{w_M}{T_l} + \frac{1 - w_M}{T_g}; \qquad \tilde{F}_T = \frac{\hat{F}_T}{T_l T_g}; \\ \tilde{F}_C = \frac{\hat{F}_C}{T} \quad \text{where } p_l = p_g + p_j,\ T_l = T_g = T. \end{gathered} \tag{79}$$

At the start of each sequential step the faster forces vanish, so there the row of the step in (77) reduces to $\tilde{F}_M = c_M \hat{F}_M$, $\tilde{F}_T = c_T \hat{F}_T$ or $\tilde{F}_C = c_C \hat{F}_C$, with $c_M = 1/T_M$, $c_T = 1/(T_l T_g)$, $c_C = 1/T$, all positive. The equilibrium of each step is therefore the vanishing of its component of $\hat{\boldsymbol{F}}$: the Laplace pressure equilibrium, the temperature equilibrium, and the equality of chemical potentials.

Jacobians. $\tilde{\boldsymbol{J}} = \partial \tilde{\boldsymbol{F}}/\partial \tilde{\boldsymbol{x}}_l$ by (51) and (53). Differentiating (77) with the product rule,

$$\tilde{J}_{ab} = \sum_c C_{ac}\,\hat{J}_{cb} + \sum_c \frac{\partial C_{ac}}{\partial \tilde{x}_{l,b}}\,\hat{F}_c. \tag{80}$$

At equilibrium $\hat{\boldsymbol{F}} = 0$ and $\tilde{\boldsymbol{J}} = \boldsymbol{C}\hat{\boldsymbol{J}}$. Away from it the two iterations take different steps toward the same root; the steps differ at second order in the residual.

Sequential solution. In components, (73) reads

$$\begin{aligned} \hat{J}^n_{MM}\,\delta\tilde{V}_l^n + \hat{J}^n_{MT}\,\delta\tilde{E}_l^n + \hat{J}^n_{MC}\,\delta\tilde{m}_l^n &= -\hat{F}^n_M, \\ \hat{J}^n_{TM}\,\delta\tilde{V}_l^n + \hat{J}^n_{TT}\,\delta\tilde{E}_l^n + \hat{J}^n_{TC}\,\delta\tilde{m}_l^n &= -\hat{F}^n_T, \\ \hat{J}^n_{CM}\,\delta\tilde{V}_l^n + \hat{J}^n_{CT}\,\delta\tilde{E}_l^n + \hat{J}^n_{CC}\,\delta\tilde{m}_l^n &= -\hat{F}^n_C. \end{aligned} \tag{81}$$

First elimination (M). The first row gives

$$\delta\tilde{V}_l^n = -\frac{\hat{F}^n_M + \hat{J}^n_{MT}\,\delta\tilde{E}_l^n + \hat{J}^n_{MC}\,\delta\tilde{m}_l^n}{\hat{J}^n_{MM}}. \tag{82}$$

Substituting (82) into the second and third rows,

$$\hat{J}'^n_{TT}\,\delta\tilde{E}_l^n + \hat{J}'^n_{TC}\,\delta\tilde{m}_l^n = -\hat{F}'^n_T, \qquad \hat{J}'^n_{CT}\,\delta\tilde{E}_l^n + \hat{J}'^n_{CC}\,\delta\tilde{m}_l^n = -\hat{F}'^n_C, \tag{83}$$

with, for $a, b \in \{T, C\}$,

$$\hat{J}'^n_{ab} \equiv \hat{J}^n_{ab} - \frac{\hat{J}^n_{aM}\hat{J}^n_{Mb}}{\hat{J}^n_{MM}}, \qquad \hat{F}'^n_a \equiv \hat{F}^n_a - \frac{\hat{J}^n_{aM}}{\hat{J}^n_{MM}}\,\hat{F}^n_M. \tag{84}$$

Second elimination (T). The first equation of (83) gives

$$\delta\tilde{E}^n_l = -\frac{\hat{F}'^n_T + \hat{J}'^n_{TC}\,\delta\tilde{m}^n_l}{\hat{J}'^n_{TT}}, \tag{85}$$

and substituting (85) into the second,

$$\hat{J}''^n_{CC}\,\delta\tilde{m}^n_l = -\hat{F}''^n_C, \qquad \hat{J}''^n_{CC} \equiv \hat{J}'^n_{CC} - \frac{\hat{J}'^n_{CT}\hat{J}'^n_{TC}}{\hat{J}'^n_{TT}}, \qquad \hat{F}''^n_C \equiv \hat{F}'^n_C - \frac{\hat{J}'^n_{CT}}{\hat{J}'^n_{TT}}\,\hat{F}'^n_T. \tag{86}$$

Solution and back substitution. (86) is a single equation, $\delta\tilde{m}^n_l = -\hat{F}''^n_C/\hat{J}''^n_{CC}$; $\delta\tilde{E}^n_l$ then follows from (85) and $\delta\tilde{V}^n_l$ from (82). For the leading 2×2 block, $\delta\tilde{m}^n_l = 0$ and the solve is $\delta\tilde{E}^n_l = -\hat{F}'^n_T/\hat{J}'^n_{TT}$, followed by (82); for the 1×1 block it is $\delta\tilde{V}^n_l = -\hat{F}^n_M/\hat{J}^n_{MM}$.

Pivots. The pivots are the coefficients that divide at each elimination: $\hat{J}^n_{MM}$ in (82), $\hat{J}'^n_{TT}$ in (85) and $\hat{J}''^n_{CC}$ in (86). They are the combinations $\hat{J}_{MM}$, $\hat{J}'_{TT}$, $\hat{J}''_{CC}$ of the entries of $\hat{\boldsymbol{J}}$, defined as in (84) and (86) at any state and evaluated at $\boldsymbol{x}^n$. Each of them is the derivative of the force of one step with respect to the transfer of that step, taken under the constraints of the step. To show this, write the differential relation (71) in components,

$$\begin{aligned} d\hat{F}_M &= \hat{J}_{MM}\,d\tilde{V}_l + \hat{J}_{MT}\,d\tilde{E}_l + \hat{J}_{MC}\,d\tilde{m}_l, \\ d\hat{F}_T &= \hat{J}_{TM}\,d\tilde{V}_l + \hat{J}_{TT}\,d\tilde{E}_l + \hat{J}_{TC}\,d\tilde{m}_l, \\ d\hat{F}_C &= \hat{J}_{CM}\,d\tilde{V}_l + \hat{J}_{CT}\,d\tilde{E}_l + \hat{J}_{CC}\,d\tilde{m}_l. \end{aligned} \tag{87}$$

At $\boldsymbol{x}^n$ this is the matrix of (81). In (81) the change of the forces is prescribed, $-\hat{\boldsymbol{F}}^n$, and the transfers are solved for. Here the opposite question is asked: how each force responds to a given transfer when the other transfers either are inactive or adjust to hold some forces fixed.

Mechanical step. Only volume is transferred, with its work along $\boldsymbol{d}_M$, at fixed masses and with no heat: $d\tilde{E}_l = d\tilde{m}_l = 0$. $\tilde{E}_l$ is the heat, not the energy: the work is carried by $\boldsymbol{d}_M = [1, -p_I, 0]^T$, so the energies still change, $dE_l = -p_I\,d\tilde{V}_l$ and $dE_g = (p_I - p_j)\,d\tilde{V}_l$ by $d\boldsymbol{x}_g = -\boldsymbol{A}\,d\boldsymbol{x}_l$, as listed in (102). The first row of (87) reduces to

$$d\hat{F}_M = \hat{J}_{MM}\,d\tilde{V}_l \quad\Longrightarrow\quad \left(\frac{\partial\hat{F}_M}{\partial\tilde{V}_l}\right)_{\tilde{E}_l,\tilde{m}_l} = \hat{J}_{MM}. \tag{88}$$

Physically, expanding the liquid lowers p_l and raises p_g, so $\hat{J}_{MM} < 0$ is expected; the sign is proven at equilibrium below.

Thermal step. Heat $d\tilde{E}_l$ is transferred at fixed masses, $d\tilde{m}_l = 0$, and the pressure equilibrium reached by the mechanical step is held, $d\hat{F}_M = 0$. Given $d\tilde{E}_l$, the unknowns are $d\tilde{V}_l$ and $d\hat{F}_T$. The first row of (87) with $d\hat{F}_M = 0$ and $d\tilde{m}_l = 0$ is

$$0 = \hat{J}_{MM}\,d\tilde{V}_l + \hat{J}_{MT}\,d\tilde{E}_l \quad\Longrightarrow\quad d\tilde{V}_l = -\frac{\hat{J}_{MT}}{\hat{J}_{MM}}\,d\tilde{E}_l, \tag{89}$$

the volume the phases must exchange to keep their pressures in equilibrium while the heat is transferred. The second row with $d\tilde{m}_l = 0$ is $d\hat{F}_T = \hat{J}_{TM}\, d\tilde{V}_l + \hat{J}_{TT}\, d\tilde{E}_l$. Substituting $d\tilde{V}_l$,

$$d\hat{F}_T = -\frac{\hat{J}_{TM}\hat{J}_{MT}}{\hat{J}_{MM}}\, d\tilde{E}_l + \hat{J}_{TT}\, d\tilde{E}_l = \Big(\hat{J}_{TT} - \frac{\hat{J}_{TM}\hat{J}_{MT}}{\hat{J}_{MM}}\Big) d\tilde{E}_l = \hat{J}'_{TT}\, d\tilde{E}_l, \tag{90}$$

where the bracket is the combination $\hat{J}'_{TT}$ of (84) with $a = b = T$, at the current state. Therefore

$$\left(\frac{\partial \hat{F}_T}{\partial \tilde{E}_l}\right)_{\hat{F}_M, \tilde{m}_l} = \hat{J}'_{TT}. \tag{91}$$

Physically, heat given by the liquid lowers T_l and raises T_g, so $\hat{J}'_{TT} < 0$ is expected.

Chemical step. Mass $d\tilde{m}_l$ is transferred, and the pressure and temperature equilibria are held, $d\hat{F}_M = d\hat{F}_T = 0$. Given $d\tilde{m}_l$, the unknowns are $d\tilde{V}_l$, $d\tilde{E}_l$ and $d\hat{F}_C$. The first row of (87) with $d\hat{F}_M = 0$ gives

$$d\tilde{V}_l = -\frac{\hat{J}_{MT}\, d\tilde{E}_l + \hat{J}_{MC}\, d\tilde{m}_l}{\hat{J}_{MM}}, \tag{92}$$

which is (82) with $\hat{F}^n_M$ replaced by $d\hat{F}_M = 0$. Substituting (92) into the second row and collecting the terms in $d\tilde{E}_l$ and in $d\tilde{m}_l$,

$$d\hat{F}_T = \Big(\hat{J}_{TT} - \frac{\hat{J}_{TM}\hat{J}_{MT}}{\hat{J}_{MM}}\Big) d\tilde{E}_l + \Big(\hat{J}_{TC} - \frac{\hat{J}_{TM}\hat{J}_{MC}}{\hat{J}_{MM}}\Big) d\tilde{m}_l = \hat{J}'_{TT}\, d\tilde{E}_l + \hat{J}'_{TC}\, d\tilde{m}_l, \tag{93}$$

and likewise into the third row,

$$d\hat{F}_C = \Big(\hat{J}_{CT} - \frac{\hat{J}_{CM}\hat{J}_{MT}}{\hat{J}_{MM}}\Big) d\tilde{E}_l + \Big(\hat{J}_{CC} - \frac{\hat{J}_{CM}\hat{J}_{MC}}{\hat{J}_{MM}}\Big) d\tilde{m}_l = \hat{J}'_{CT}\, d\tilde{E}_l + \hat{J}'_{CC}\, d\tilde{m}_l, \tag{94}$$

with the combinations of (84) at the current state. These are the rows (83) with the increments $d\hat{F}_T$ and $d\hat{F}_C$ in place of $-\hat{F}'^n_T$ and $-\hat{F}'^n_C$. Holding $d\hat{F}_T = 0$ in the first,

$$d\tilde{E}_l = -\frac{\hat{J}'_{TC}}{\hat{J}'_{TT}}\, d\tilde{m}_l, \tag{95}$$

the heat that must accompany the mass to keep the temperatures in equilibrium, which is (85) with $\hat{F}'^n_T$ replaced by $d\hat{F}_T = 0$. Substituting into the second,

$$d\hat{F}_C = -\frac{\hat{J}'_{CT}\hat{J}'_{TC}}{\hat{J}'_{TT}}\, d\tilde{m}_l + \hat{J}'_{CC}\, d\tilde{m}_l = \Big(\hat{J}'_{CC} - \frac{\hat{J}'_{CT}\hat{J}'_{TC}}{\hat{J}'_{TT}}\Big) d\tilde{m}_l = \hat{J}''_{CC}\, d\tilde{m}_l, \tag{96}$$

where the bracket is the combination $\hat{J}''_{CC}$ of (86) at the current state. Therefore

$$\left(\frac{\partial \hat{F}_C}{\partial \tilde{m}_l}\right)_{\hat{F}_M, \hat{F}_T} = \hat{J}''_{CC}. \tag{97}$$

Physically, mass given by the liquid lowers g_l relative to g_v, so $\hat{J}''_{CC} < 0$ is expected.

Elimination and constraints. The three derivations repeat the algebra of (82)–(86): eliminating $\delta\tilde{V}_l$ with the first row is what holding $\hat{F}_M$ does, and eliminating $\delta\tilde{E}_l$ with the reduced second row

is what holding $\hat{F}_T$ does. The pivots are therefore the derivatives of the forces under exactly the constraints of the sequential steps. At the start state of each step, these are the derivatives that the sequential relaxation below computes from the entries of $\hat{\boldsymbol{J}}$: (115) for the mechanical step at any state, (119) for the thermal step at $\hat{F}_M = 0$, and (127) for the chemical step at $\hat{F}_M = \hat{F}_T = 0$. At $\hat{F}_M = 0$ the interface pressure, which lies between $p_g + p_j$ and p_l, equals p_l, so the heat $d\tilde{E}_l = -(dE_l + p_I\, dV_l)$ is the heat given by the liquid when each phase works at its own pressure. With every transfer counted in the direction its force drives it, each force decreases along its own transfer, and the stiffnesses of the steps are defined as the magnitudes of the three derivatives (positivity is shown below),

$$\left(\frac{\partial \hat{F}_M}{\partial \tilde{V}_l}\right)_{\tilde{E}_l,\tilde{m}_l} = \hat{J}_{MM} = -\mathcal{K}^M, \quad \left(\frac{\partial \hat{F}_T}{\partial \tilde{E}_l}\right)_{\hat{F}_M,\tilde{m}_l} = \hat{J}'_{TT} = -\mathcal{K}^T, \quad \left(\frac{\partial \hat{F}_C}{\partial \tilde{m}_l}\right)_{\hat{F}_M,\hat{F}_T} = \hat{J}''_{CC} = -\mathcal{K}^C. \tag{98}$$

Positivity. At equilibrium $\hat{F}_M = 0$, so p_I, which lies between $p_g + p_j$ and p_l, equals p_l, and $C_{MT} = 0$ in (77): $\boldsymbol{C}$ is lower triangular. By (80), $\tilde{\boldsymbol{J}} = \boldsymbol{C}\hat{\boldsymbol{J}}$, and for a lower-triangular $\boldsymbol{C}$ the leading $k \times k$ block of the product is the product of the leading blocks, $\tilde{\boldsymbol{J}}_k = \boldsymbol{C}_k\hat{\boldsymbol{J}}_k$. Taking determinants, $\det \tilde{\boldsymbol{J}}_k = \left(\prod_{i\le k} C_{ii}\right) \det \hat{\boldsymbol{J}}_k$. The elimination makes each leading determinant the product of the first k pivots, so the ratio of consecutive leading determinants is the k-th pivot, and the pivots of $\tilde{\boldsymbol{J}}$ are C_{ii} times those of $\hat{\boldsymbol{J}}$. $\tilde{\boldsymbol{J}} = -\tilde{\boldsymbol{K}}$ with $\tilde{\boldsymbol{K}}$ positive definite, so its leading determinants are positive and the pivots of $\tilde{\boldsymbol{J}}$ are $-s_i$ with

$$s_M = \tilde{K}_{MM} > 0, \qquad s_T = \frac{\det \tilde{\boldsymbol{K}}_{MT}}{\tilde{K}_{MM}} > 0, \qquad s_C = \frac{\det \tilde{\boldsymbol{K}}}{\det \tilde{\boldsymbol{K}}_{MT}} > 0, \tag{99}$$

where $\tilde{\boldsymbol{K}}_{MT}$ is the leading 2×2 block. With $C_{MM} = 1/T$, $C_{TT} = 1/T^2$, $C_{CC} = 1/T$ and the pivots $-\mathcal{K}^M$, $-\mathcal{K}^T$, $-\mathcal{K}^C$ of $\hat{\boldsymbol{J}}$,

$$\begin{gathered} -s_M = C_{MM}(-\mathcal{K}^M), \qquad -s_T = C_{TT}(-\mathcal{K}^T), \qquad -s_C = C_{CC}(-\mathcal{K}^C) \\ \Longrightarrow \quad \mathcal{K}^M = T\, s_M, \qquad \mathcal{K}^T = T^2 s_T, \qquad \mathcal{K}^C = T\, s_C, \end{gathered} \tag{100}$$

all positive at equilibrium, and by continuity near it. Away from equilibrium $\mathcal{K}^M$ of (115) depends on p_I. For the ideal gas, the numerator of its gas term is $K_{s,g}+\Gamma_g(p_I-p_j-p_g) = p_g+(\gamma_g-1)(p_I-p_j)$, which is negative when $p_I - p_j < -p_g/(\gamma_g - 1)$, i.e. when p_I is taken close to the pressure of a liquid in tension; with a small gas volume $\mathcal{K}^M$ is then negative. The pivot of the entropy force, $\tilde{K}_{MM} = \boldsymbol{d}_M^T \boldsymbol{K} \boldsymbol{d}_M$, is positive at every state by the definiteness of $\boldsymbol{K}$; the two differ by the term $\hat{F}_M\, \partial(1/T_M)/\partial \tilde{V}_l$ of (80).

Sequential steps. Each elimination is one relaxation step:

1. mechanical: (82) with $\delta\tilde{E}_l^n = \delta\tilde{m}_l^n = 0$, after which $\hat{F}_M^{n+1} = 0$ to first order;
2. thermal, from a state with $\hat{F}_M^n = 0$: there $\hat{F}_T'^n = \hat{F}_T^n$ by (84), and (85) with $\delta\tilde{m}_l^n = 0$ gives $\delta\tilde{E}_l^n$, with $\delta\tilde{V}_l^n = -(\hat{J}_{MT}^n/\hat{J}_{MM}^n)\,\delta\tilde{E}_l^n$ holding $\hat{F}_M = 0$; after it $\hat{F}_M^{n+1} = \hat{F}_T^{n+1} = 0$ to first order;
3. chemical, from a state with $\hat{F}_M^n = \hat{F}_T^n = 0$: there $\hat{F}_C''^n = \hat{F}_C^n$, and (86) gives $\delta\tilde{m}_l^n$, with $\delta\tilde{V}_l^n$ and $\delta\tilde{E}_l^n$ from (82) and (85) holding $\hat{F}_M = \hat{F}_T = 0$; after it $\hat{\boldsymbol{F}}^{n+1} = 0$ to first order.

With (98) evaluated at the start state $\boldsymbol{x}^n$ of each step, the transfers of the three steps are

$$\delta\tilde{V}_l^n = \frac{\hat{F}_M^n}{\mathcal{K}^{M,n}}, \qquad \delta\tilde{E}_l^n = \frac{\hat{F}_T^n}{\mathcal{K}^{T,n}}, \qquad \delta\tilde{m}_l^n = \frac{\hat{F}_C^n}{\mathcal{K}^{C,n}}. \tag{101}$$

With $\hat{\boldsymbol{J}}$ frozen at $\boldsymbol{x}^n$, each step leaves the linearized faster forces at zero and after the third all three vanish, so the three steps add up to the Newton step (73). In the sequential relaxation each step is instead evaluated at its own start state, where the faster forces vanish: its right-hand side is the component of $\hat{\boldsymbol{F}}$ of the step at that state, and its pivot is evaluated there. A step is then the Newton step of a constrained entropy maximum: over $\mathrm{span}\{\boldsymbol{d}_M\}$ (mechanical), over $\{\delta m_l = 0\}$ (thermo-mechanical), and over all exchanges (full equilibrium). Iterating each stage, as described at the end of the sequential relaxation, reaches these maxima exactly and ends at $\boldsymbol{F} = 0$.

1.1.4 Sequential relaxation

The three steps are the eliminations (82)–(86) of the Newton system (73), each evaluated at its own start state. The pre-relaxation, post-mechanical, post-thermal and post-chemical states are marked by the superscripts 0, $*$, $**$ and $\otimes$. The start state of each step (0, $*$, $**$) is the state $\boldsymbol{x}^n$ of (101) for that step, and the forces of the step are the components of $\hat{\boldsymbol{F}}$ evaluated there. Finite increments over a step are written with δ, differential relations with d. The steps need the entries of $\hat{\boldsymbol{J}}$, which are derived first.

Jacobian in the transfer coordinates. By (71), the column $b \in \{M, T, C\}$ of $\hat{\boldsymbol{J}}$ is the change of $\hat{\boldsymbol{F}}$ per unit transfer along $\boldsymbol{d}_b$, for which the liquid changes by $d\boldsymbol{x}_l = \boldsymbol{d}_b$ and the gas by $d\boldsymbol{x}_g = -\boldsymbol{A}\boldsymbol{d}_b$:

$$\begin{array}{lll} & d\boldsymbol{x}_l = [dV_l,\, dE_l,\, dm_l]^T & d\boldsymbol{x}_g = [dV_g,\, dE_g,\, dm_v]^T \\ b = M: & [1,\, -p_I,\, 0]^T & [-1,\, p_I - p_j,\, 0]^T \\ b = T: & [0,\, -1,\, 0]^T & [0,\, 1,\, 0]^T \\ b = C: & [-v_l,\, -e_l,\, -1]^T & [v_l,\, e_l + p_j v_l,\, 1]^T \end{array} \tag{102}$$

per unit $\delta\tilde{V}_l$, $\delta\tilde{E}_l$ and $\delta\tilde{m}_l$, respectively. Write ∂_b for the derivative along $\boldsymbol{d}_b$ with the other two transfer coordinates fixed,

$$\partial_M \equiv \left(\frac{\partial}{\partial \tilde{V}_l}\right)_{\tilde{E}_l, \tilde{m}_l}, \qquad \partial_T \equiv \left(\frac{\partial}{\partial \tilde{E}_l}\right)_{\tilde{V}_l, \tilde{m}_l}, \qquad \partial_C \equiv \left(\frac{\partial}{\partial \tilde{m}_l}\right)_{\tilde{V}_l, \tilde{E}_l}. \tag{103}$$

By (67), the entries of $\hat{\boldsymbol{J}}$ are the differences of the phase changes,

$$\hat{J}_{Mb} = \partial_b p_l - \partial_b p_g, \qquad \hat{J}_{Tb} = \partial_b T_l - \partial_b T_g, \qquad \hat{J}_{Cb} = \partial_b g_l - \partial_b g_v. \tag{104}$$

Phase at fixed mass. The Grüneisen coefficient, the specific heat at constant volume and the isentropic bulk modulus are

$$\Gamma \equiv v\left(\frac{\partial p}{\partial e}\right)_v, \qquad c_v \equiv \left(\frac{\partial e}{\partial T}\right)_v, \qquad K_s \equiv -v\left(\frac{\partial p}{\partial v}\right)_s. \tag{105}$$

By (1),

$$\left(\frac{\partial e}{\partial s}\right)_v = T, \qquad \left(\frac{\partial s}{\partial v}\right)_e = \frac{p}{T}, \qquad \left(\frac{\partial T}{\partial v}\right)_s = -\left(\frac{\partial p}{\partial s}\right)_v, \tag{106}$$

the last being a Maxwell relation. Hence $(\partial p/\partial s)_v = (\partial p/\partial e)_v(\partial e/\partial s)_v = \Gamma T/v$ and $(\partial T/\partial s)_v = T/c_v$, and

$$\begin{aligned} \left(\frac{\partial p}{\partial v}\right)_e &= \left(\frac{\partial p}{\partial v}\right)_s + \left(\frac{\partial p}{\partial s}\right)_v \left(\frac{\partial s}{\partial v}\right)_e = \frac{\Gamma p - K_s}{v}, \\ \left(\frac{\partial T}{\partial v}\right)_e &= \left(\frac{\partial T}{\partial v}\right)_s + \left(\frac{\partial T}{\partial s}\right)_v \left(\frac{\partial s}{\partial v}\right)_e = \frac{p}{c_v} - \frac{\Gamma T}{v}. \end{aligned} \tag{107}$$

At fixed mass $v = V/m$ and $e = E/m$, so with $C_v \equiv mc_v$, and with (4),

$$dp = \frac{(\Gamma p - K_s)\,dV + \Gamma\,dE}{V}, \qquad dT = \Big(\frac{p}{C_v} - \frac{\Gamma T}{V}\Big)dV + \frac{dE}{C_v}, \qquad dg = v\,dp - s\,dT. \tag{108}$$

For the gas, (108) holds at fixed composition with the mixture coefficients Γ_g, $K_{s,g}$, $C_{v,g}$. At fixed composition p_v/p_g is fixed, so the vapor chemical potential $g_v(p_v, T_g)$ changes by $dg_v = \bar{v}_v\,dp_g - \bar{s}_v\,dT_g$, with $\bar{v}_v \equiv v_v p_v/p_g = R_v T_g/p_g$ and $\bar{s}_v \equiv s_v(p_v, T_g)$.

Gas mass. Along $\boldsymbol{d}_C$ the gas also receives vapor at fixed m_a. For the ideal-gas mixture, $T_g = (E_g - m_v q_v - m_a q_a)/C_{v,g}$ and $p_g = (m_v R_v + m_a R_a)T_g/V_g$, with $C_{v,g} = m_v c_{v,v} + m_a c_{v,a}$ and $\Gamma_g = (m_v R_v + m_a R_a)/C_{v,g}$, so

$$\Big(\frac{\partial T_g}{\partial m_v}\Big)_{V_g,E_g,m_a} = -\frac{e_v}{C_{v,g}}, \qquad \Big(\frac{\partial p_g}{\partial m_v}\Big)_{V_g,E_g,m_a} = \frac{R_v T_g - \Gamma_g e_v}{V_g}, \qquad e_v \equiv c_{v,v}T_g + q_v. \tag{109}$$

With the vapor mole fraction $x_v \equiv p_v/p_g = m_v R_v/(m_v R_v + m_a R_a)$, $dp_v = x_v\,dp_g + p_g\,dx_v$ and $dx_v = x_v(1 - x_v)\,dm_v/m_v$ at fixed m_a, so

$$dg_v = \bar{v}_v\,dp_g - \bar{s}_v\,dT_g + \mathcal{K}_{CC}\,dm_v, \qquad \mathcal{K}_{CC} \equiv \frac{R_v T_g(1 - x_v)}{m_v}. \tag{110}$$

Entries. Substituting (102) into (108), (109) and (110): along $\boldsymbol{d}_M$,

$$\begin{aligned} \partial_M p_l &= -\frac{K_{s,l} + \Gamma_l(p_I - p_l)}{V_l}, & \partial_M T_l &= \frac{p_l - p_I}{C_{v,l}} - \frac{\Gamma_l T_l}{V_l}, \\ \partial_M p_g &= \frac{K_{s,g} + \Gamma_g(p_I - p_j - p_g)}{V_g}, & \partial_M T_g &= \frac{p_I - p_j - p_g}{C_{v,g}} + \frac{\Gamma_g T_g}{V_g}; \end{aligned} \tag{111}$$

along $\boldsymbol{d}_T$,

$$\partial_T p_l = -\frac{\Gamma_l}{V_l}, \qquad \partial_T T_l = -\frac{1}{C_{v,l}}, \qquad \partial_T p_g = \frac{\Gamma_g}{V_g}, \qquad \partial_T T_g = \frac{1}{C_{v,g}}; \tag{112}$$

along $\boldsymbol{d}_C$ the liquid gives mass at its own specific volume and energy, so its intensive state does not change, $\partial_C p_l = \partial_C T_l = \partial_C g_l = 0$, and

$$\partial_C p_g = \frac{\Gamma_g\big(e_l + (p_g + p_j)v_l - e_v\big) - K_{s,g}v_l + R_v T_g}{V_g}, \qquad \partial_C T_g = \frac{e_l + (p_g + p_j)v_l - e_v}{C_{v,g}} - \frac{\Gamma_g T_g v_l}{V_g}. \tag{113}$$

For the chemical potentials, the masses are fixed along $\boldsymbol{d}_M$ and $\boldsymbol{d}_T$, and the vapor mass changes along $\boldsymbol{d}_C$:

$$\partial_b g_l = v_l\,\partial_b p_l - s_l\,\partial_b T_l, \qquad \partial_b g_v = \bar{v}_v\,\partial_b p_g - \bar{s}_v\,\partial_b T_g + \mathcal{K}_{CC}\,\delta_{bC}. \tag{114}$$

(104) with (111)–(114) gives all entries of $\hat{\boldsymbol{J}}$ at any state.

Mechanical step. From the start state 0, the mechanical step is (82) with $\delta\tilde{E}_l = \delta\tilde{m}_l = 0$, i.e. $\hat{J}_{MM}\,\delta\tilde{V}_l = -\hat{F}^0_M$. By (104) and (111), $\hat{J}_{MM} = -\mathcal{K}^M$ with

$$\mathcal{K}^M = \frac{K_{s,l} + \Gamma_l(p_I - p_l)}{V_l} + \frac{K_{s,g} + \Gamma_g(p_I - p_j - p_g)}{V_g}, \qquad \delta\tilde{V}_l = \frac{\hat{F}^0_M}{\mathcal{K}^M}, \tag{115}$$

all evaluated at 0. By (102), the extensive increments are $\delta V_l = -\delta V_g = \delta\tilde{V}_l$, $\delta E_l = -p_I\,\delta\tilde{V}_l$, $\delta E_g = (p_I - p_j)\,\delta\tilde{V}_l$, $\delta m_l = \delta m_v = 0$, and the intensive increments are

$$\delta p^M_k = \partial_M p_k\,\delta\tilde{V}_l, \qquad \delta T^M_k = \partial_M T_k\,\delta\tilde{V}_l, \qquad k \in \{l, g\}, \tag{116}$$

with the coefficients (111).

End state. Along $\boldsymbol{d}_M$ with p_I held, by (49) and (65), $dS = \tilde{F}_M\, d\tilde{V}_l$ with $\tilde{F}_M = (p_l - p_I)/T_l + (p_I - p_g - p_j)/T_g$. While p_I lies between $p_g + p_j$ and p_l, (79) gives $\tilde{F}_M = \hat{F}_M/T_M$ with $1/T_M > 0$, and S increases as the liquid expands for $\hat{F}_M > 0$ and contracts for $\hat{F}_M < 0$. At the Laplace pressure equilibrium of the path, $p_l = p_g + p_j \equiv p^*$, however,

$$\tilde{F}_M = (p^* - p_I)\Big(\frac{1}{T_l} - \frac{1}{T_g}\Big), \tag{117}$$

which for $T_l \neq T_g$ vanishes only if $p_I = p^*$. For any other constant p_I, the pressures pass p_I before they equalize, p_I leaves the interval between $p_g + p_j$ and p_l, and the maximum of S on the path lies before the pressure equilibrium. The instantaneous mechanical step therefore takes $p_I = p^*$, the pressure of its own end state. Then, where p_l decreases and p_g increases along $\boldsymbol{d}_M$, i.e. where both numerators of $\mathcal{K}^M$ in (115) are positive, $p_l > p^* > p_g + p_j$ before the end state for $\hat{F}^0_M > 0$ (and the reverse for $\hat{F}^0_M < 0$): p^* lies between the two pressures at every state of the path, $\tilde{F}_M = \hat{F}_M/T_M$ keeps the sign of $\hat{F}^0_M$ up to the end state and changes it there, and the end state is the unique maximum of S on its path. For a stiffened-gas or NASG liquid and the ideal-gas mixture, the numerators are $\big(p_l + \gamma_l p_\infty + (\gamma_l - 1)p_I\big)/(1 - b\rho_l)$ and $p_g + \Gamma_g(p_I - p_j)$ (Algorithm, step 2), both positive for $p_I = p^*$, since $p^* > -p_\infty$ and $p^* - p_j = p^*_g > 0$. Pressure equilibrium is one condition; the split of the energy is set by the work at p_I in $\boldsymbol{d}_M$, here at the end pressure, so the end state depends on no other closure. The linear step above, with p_I at 0, agrees with it to first order, since $p^0_I - p^* = \mathcal{O}(\hat{F}^0_M)$.

Choice of p_I for the instantaneous step. Every constant p_I gives a path along $\boldsymbol{d}_M$ that ends in pressure equilibrium, $\hat{F}_M = 0$, and conserves $E_l + E_g + p_j V_l$; the paths differ only in the split of the energy, through the work $p_I\, \delta\tilde{V}_l$. The equivalence of $\hat{F}_M = 0$ and $\tilde{F}_M = 0$ in (79) rests on (78), i.e. on p_I lying between the pressures of the state at which the forces are evaluated. A p_I evaluated at the start state, such as the impedance-weighted value of the Algorithm, satisfies this at the start but not at the end state, where both pressures equal p^*, so there (117) is non-zero: the state of pressure equilibrium is not the maximum of S on its path, and its entropy differs from that maximum at second order in $\hat{F}^0_M$. $p_I = p^*$ satisfies (78) at every state of its path, so its end state is at once the pressure equilibrium and the maximum of S on the path, which is the definition of the end states of all three stages; it needs no interface-pressure closure, and for a NASG liquid it has the closed form below. It is therefore the p_I of the instantaneous mechanical step. p^* is not known before the step: for a NASG liquid the end state has the closed form below, and for a general equation of state it is the root of the system (130), solved by Newton's method in the end state and p^* together. The finite-rate step keeps p_I at its start state, between the current pressures, where (139) holds.

Thermal step. From the start state $*$, where $\hat{F}^*_M = 0$ and therefore $p_I = p_l = p_g + p_j$, the thermal step is (85) with $\delta\tilde{m}_l = 0$. At this state (104) with (111) and (112) gives

$$\begin{aligned} \hat{J}_{MM} &= -\Big(\frac{K_{s,l}}{V_l} + \frac{K_{s,g}}{V_g}\Big), & \hat{J}_{MT} &= -\Big(\frac{\Gamma_l}{V_l} + \frac{\Gamma_g}{V_g}\Big), \\ \hat{J}_{TM} &= -\Big(\frac{\Gamma_l T_l}{V_l} + \frac{\Gamma_g T_g}{V_g}\Big), & \hat{J}_{TT} &= -\Big(\frac{1}{C_{v,l}} + \frac{1}{C_{v,g}}\Big). \end{aligned} \tag{118}$$

With $\hat{F}^*_M = 0$, $\hat{F}'_T = \hat{F}^*_T$ by (84), and $\hat{J}'_{TT} = -\mathcal{K}^T$ with

$$\mathcal{K}^T = \frac{1}{C_{v,l}} + \frac{1}{C_{v,g}} - \frac{\big(\Gamma_l T_l/V_l + \Gamma_g T_g/V_g\big)\big(\Gamma_l/V_l + \Gamma_g/V_g\big)}{K_{s,l}/V_l + K_{s,g}/V_g}, \qquad \delta\tilde{E}_l = \frac{\hat{F}^*_T}{\mathcal{K}^T}, \tag{119}$$

all evaluated at $*$. The volume that holds $\hat{F}_M = 0$ follows from (82) with a zero force term,

$$\delta\tilde{V}_l = -\frac{\hat{J}_{MT}}{\hat{J}_{MM}}\,\delta\tilde{E}_l = -\frac{\Gamma_l/V_l + \Gamma_g/V_g}{K_{s,l}/V_l + K_{s,g}/V_g}\,\delta\tilde{E}_l. \tag{120}$$

By (102), the extensive increments are $\delta V_l = -\delta V_g = \delta\tilde{V}_l$, $\delta E_l = -p_l\,\delta\tilde{V}_l - \delta\tilde{E}_l$, $\delta E_g = p_g\,\delta\tilde{V}_l + \delta\tilde{E}_l$, $\delta m_l = \delta m_v = 0$: each phase works at its own pressure and the liquid gives the heat $\delta\tilde{E}_l$. The intensive increments are $\delta z_k^T = \partial_M z_k\,\delta\tilde{V}_l + \partial_T z_k\,\delta\tilde{E}_l$, i.e. by (111) and (112) at $p_I = p_l$,

$$\begin{gathered}\delta p^T = -\frac{K_{s,l}}{V_l}\,\delta\tilde{V}_l - \frac{\Gamma_l}{V_l}\,\delta\tilde{E}_l = \frac{K_{s,g}}{V_g}\,\delta\tilde{V}_l + \frac{\Gamma_g}{V_g}\,\delta\tilde{E}_l,\\ \delta T_l^T = -\frac{\Gamma_l T_l}{V_l}\,\delta\tilde{V}_l - \frac{\delta\tilde{E}_l}{C_{v,l}}, \qquad \delta T_g^T = \frac{\Gamma_g T_g}{V_g}\,\delta\tilde{V}_l + \frac{\delta\tilde{E}_l}{C_{v,g}},\end{gathered} \tag{121}$$

where the two forms of δp^T agree because $\hat{J}_{MM}\,\delta\tilde{V}_l + \hat{J}_{MT}\,\delta\tilde{E}_l = 0$. Per unit $\delta\tilde{E}_l$, these are the derivatives $(\partial p/\partial\tilde{E}_l)_{\hat{F}_M,\tilde{m}_l}$ and $(\partial T_k/\partial\tilde{E}_l)_{\hat{F}_M,\tilde{m}_l}$, and $\mathcal{K}^T = (\partial T_g/\partial\tilde{E}_l)_{\hat{F}_M,\tilde{m}_l} - (\partial T_l/\partial\tilde{E}_l)_{\hat{F}_M,\tilde{m}_l}$.

End state. Along the step $\hat{F}_M = 0$ is held, so $\tilde{F}_M = 0$ by (79), and by (49)

$$dS = \tilde{F}_T\,d\tilde{E}_l = \frac{\hat{F}_T}{T_l T_g}\,d\tilde{E}_l. \tag{122}$$

S is stationary only at $\hat{F}_T = T_l - T_g = 0$, and there $(\partial\tilde{F}_T/\partial\tilde{E}_l)_{\hat{F}_M,\tilde{m}_l} = -\mathcal{K}^T/(T_l T_g) < 0$, so the end state is the maximum of S at fixed masses: $\hat{F}_M = \hat{F}_T = 0$, two conditions for the two unknowns V_l and E_l, independent of p_I.

Chemical step. From the start state $**$, where $\hat{F}_M^{**} = \hat{F}_T^{**} = 0$, so that $p_I = p_l = p_g + p_j$, $T_l = T_g = T$ and $e_l + (p_g + p_j)v_l = h_l$, the chemical step is (86), with $\delta\tilde{V}_l$ and $\delta\tilde{E}_l$ from (82) and (85). The block of the rows and columns M, T is (118) with $T_l = T_g = T$, and the column C is, by (104), (113) and (114),

$$\begin{gathered}\hat{J}_{MC} = -\frac{\Gamma_g(h_l - e_v) - K_{s,g}v_l + R_v T}{V_g}, \qquad \hat{J}_{TC} = -\frac{h_l - e_v}{C_{v,g}} + \frac{\Gamma_g T v_l}{V_g},\\ \hat{J}_{CC} = \bar{v}_v\hat{J}_{MC} - \bar{s}_v\hat{J}_{TC} - \mathcal{K}_{CC}.\end{gathered} \tag{123}$$

With $\hat{F}_M^{**} = \hat{F}_T^{**} = 0$, $\hat{F}_C'' = \hat{F}_C^{**}$ by (84) and (86), so $\delta\tilde{m}_l = \hat{F}_C^{**}/\mathcal{K}^C$ with $\mathcal{K}^C \equiv -\hat{J}_{CC}''$. The heat and volume that accompany the mass follow from (85) and (82) with zero force terms; with $\hat{J}_{MM} = -\mathcal{K}^M$ and $\hat{J}_{TT}' = -\mathcal{K}^T$ of (115) and (119) at $**$,

$$\delta\tilde{E}_l = \frac{\hat{J}_{TC}'}{\mathcal{K}^T}\,\delta\tilde{m}_l, \quad \hat{J}_{TC}' = \hat{J}_{TC} + \frac{\hat{J}_{TM}\hat{J}_{MC}}{\mathcal{K}^M}, \qquad \delta\tilde{V}_l = \frac{\hat{J}_{MT}\,\delta\tilde{E}_l + \hat{J}_{MC}\,\delta\tilde{m}_l}{\mathcal{K}^M}. \tag{124}$$

The liquid changes only through $\delta\tilde{V}_l$ and $\delta\tilde{E}_l$, since $\partial_C z_l = 0$, and the gas has the same pressure and temperature increments because $\hat{F}_M$ and $\hat{F}_T$ are held. By (111) and (112) at $p_I = p_l$,

$$\delta p^C = -\frac{K_{s,l}}{V_l}\,\delta\tilde{V}_l - \frac{\Gamma_l}{V_l}\,\delta\tilde{E}_l \equiv -\Pi\,\delta\tilde{m}_l, \qquad \delta T^C = -\frac{\Gamma_l T}{V_l}\,\delta\tilde{V}_l - \frac{\delta\tilde{E}_l}{C_{v,l}} \equiv -\Theta\,\delta\tilde{m}_l, \tag{125}$$

which define Π and Θ, the changes of p and T per unit mass received by the liquid.

Pivot. By the derivation of the pivots (98), $\hat{J}''_{CC}\,\delta\tilde{m}_l$ is the change of $\hat{F}_C = g_l - g_v$ along the step. By (114), the liquid potential changes by $v_l\,\delta p^C - s_l\,\delta T^C$, and the vapor potential, summed over the three transfers, by $\bar{v}_v\,\delta p^C - \bar{s}_v\,\delta T^C + \mathcal{K}_{CC}\,\delta\tilde{m}_l$. With $\Delta v \equiv v_l - \bar{v}_v$ and $\Delta s \equiv s_l - \bar{s}_v$,

$$\hat{J}''_{CC}\,\delta\tilde{m}_l = \Delta v\,\delta p^C - \Delta s\,\delta T^C - \mathcal{K}_{CC}\,\delta\tilde{m}_l, \tag{126}$$

and by (125)

$$\mathcal{K}^C = -\hat{J}''_{CC} = \Delta v\,\Pi - \Delta s\,\Theta + \mathcal{K}_{CC}, \qquad \delta\tilde{m}_l = \frac{\hat{F}^{**}_C}{\mathcal{K}^C}, \tag{127}$$

all evaluated at $**$. By (102), the extensive increments are $\delta m_l = -\delta m_v = -\delta\tilde{m}_l$, $\delta V_l = -\delta V_g = \delta\tilde{V}_l - v_l\,\delta\tilde{m}_l$, $\delta E_l = -p_l\,\delta\tilde{V}_l - \delta\tilde{E}_l - e_l\,\delta\tilde{m}_l$ and $\delta E_g = -\delta E_l - p_j\,\delta V_l$.

End state. Along the step $\hat{F}_M = \hat{F}_T = 0$ are held, so $\tilde{F}_M = \tilde{F}_T = 0$ and $\tilde{F}_C = \hat{F}_C/T$ by (79), and $dS = \tilde{F}_C\,d\tilde{m}_l = \hat{F}_C\,d\tilde{m}_l/T$: the volume and energy that accompany the mass generate no entropy, and the result does not depend on $\boldsymbol{d}_C$. S is stationary only at $\hat{F}_C = g_l - g_v = 0$, with g_v at the vapor partial pressure, and there $(\partial\tilde{F}_C/\partial\tilde{m}_l)_{\hat{F}_M,\hat{F}_T} = -\mathcal{K}^C/T < 0$, so the end state is the maximum of S over all exchanges, $\boldsymbol{F} = 0$.

Dilute vapor. $\mathcal{K}_{CC} = R_v T_g(1 - x_v)/m_v$ is singular as $m_v \to 0$, and $\hat{F}_C$ contains $-R_v T_g \ln x_v$ through $g_v(p_v, T_g)$: in the dilute limit $\hat{F}_C$ is linear in $\ln m_v$, not in m_v. In the iteration of the chemical stage the Newton unknown of the mass is therefore $u = \ln m_v$: with $\delta u = \delta\tilde{m}_l/m_v$ from (86), the new vapor mass is $m_v e^{\delta u}$, the transferred mass $m_v(e^{\delta u} - 1)$ agrees with $\delta\tilde{m}_l$ to first order, and $\delta\tilde{E}_l$ and $\delta\tilde{V}_l$ follow from (85) and (82) with this mass. The vapor mass is set to $m_v e^{\delta u}$ itself, not computed as $m_v + m_v(e^{\delta u} - 1)$, which rounds to zero once $e^{\delta u}$ is below the machine precision. The root is unchanged and m_v stays positive. A dry gas ($m_v = 0$, $\hat{F}_C = +\infty$) is first given a vapor mass of negligible mole fraction along $\boldsymbol{d}_C$.

Summary. From its own start state, each step transfers $\delta\tilde{V}_l = \hat{F}^0_M/\mathcal{K}^M$, $\delta\tilde{E}_l = \hat{F}^*_T/\mathcal{K}^T$, $\delta\tilde{m}_l = \hat{F}^{**}_C/\mathcal{K}^C$ with the faster equilibria held and each stiffness evaluated at the start state of its step. The stiffnesses $\mathcal{K}^M$, $\mathcal{K}^T$, $\mathcal{K}^C$ derived above from the entries of $\hat{\boldsymbol{J}}$ are the pivots (98) of the Newton system (73) at the start state of each step, so each step is an elimination step of (73). One pass of the three steps reaches the equilibria of the steps to first order; iterating the stages as below reaches them exactly, and the last stage ends at the full equilibrium $\boldsymbol{F} = 0$ of (25).

Update of pressure and temperature. The transfers of a step give the new extensive state exactly, $\boldsymbol{x}_l^{n+1} = \boldsymbol{x}_l^n + \boldsymbol{D}^n\,\delta\tilde{\boldsymbol{x}}_l^n$ and $\boldsymbol{x}_g^{n+1} = \boldsymbol{x}_g^n - \boldsymbol{A}\boldsymbol{D}^n\,\delta\tilde{\boldsymbol{x}}_l^n$ by (54). The intensive state (p_k, T_k, g_k) at the start of the next step can be obtained in three ways.

1. *Linear update*, as in the steps above: p_k and T_k are advanced by the linear increments of the step (δp_k^M, δT_k^M, δp^T, δT_k^T, δp^C, δT^C). The equilibria of the step then hold exactly in the updated values, e.g. $p_l^* - p_g^* = p_j$, so the next step starts with its faster forces exactly zero, as its derivation assumes. The updated values are, however, not the equation-of-state values at $\boldsymbol{x}^{n+1}$; they differ from them by $\mathcal{O}(\|\hat{\boldsymbol{F}}^n\|^2)$, so the intensive and extensive states are consistent only to first order.

2. *Equation-of-state update*: p_k, T_k and g_k are evaluated from the equations of state at $\boldsymbol{x}^{n+1}$. The state is then consistent, but the equilibria of the step hold only to first order, and the next step starts with faster forces of order $\mathcal{O}(\|\hat{\boldsymbol{F}}^n\|^2)$ instead of zero. These residuals enter the elimination through the full right-hand sides $\hat{F}'^n_T$ and $\hat{F}'^n_C$ of (84) and (86), and through

the term $\hat{F}_M^n$ of (82); keeping them, instead of $\hat{F}_T^n$ and $\hat{F}_C^n$ alone, makes the next step a Newton step that also corrects the faster forces.

3. *Mixture-energy update*, Pelanti's closure [1, eqs. (59), (70), (82)]: V_l and the masses are taken from $\boldsymbol{x}^{n+1}$, and the energy is split between the phases so that $\hat{F}_M = 0$ exactly, with $E_l + E_g + p_j V_l$ unchanged. This adds to the transfer of the step a heat $\delta\tilde{E}_l^c$ along $\boldsymbol{d}_T$, the root of $\hat{F}_M(\boldsymbol{x}_l^{n+1} + \boldsymbol{d}_T\,\delta\tilde{E}_l^c) = 0$, which is of order $\|\hat{\boldsymbol{F}}^n\|^2$ and changes neither V_l nor the masses. The state is consistent and in exact pressure equilibrium, but the temperature (and chemical) equilibria of the step hold only to first order. Because the correction is a heat exchanged at frozen volumes, the $\mathcal{O}(\|\hat{\boldsymbol{F}}^n\|^2)$ energy error of the linear step changes mostly the temperature of the phase with the smaller heat capacity.

The three options give the same transfers to first order. Option 1 is a single linearized pass; option 2 is required for the iteration below; option 3 is Pelanti's one-pass procedure.

Iteration of the stages. A single pass takes one Newton step per stage and reaches the relaxed states to first order. The relaxed states themselves are reached by iterating (73) with option 2, $n = 0, 1, \ldots$ from the start state $\boldsymbol{x}^0$ of the relaxation, with $\hat{\boldsymbol{J}}^n$ evaluated with $\boldsymbol{G}_k$ and $\boldsymbol{K}_k$ at $\boldsymbol{x}^n$ and $\boldsymbol{D}$ held at $\boldsymbol{x}^0$, with p_I of (78) at $\boldsymbol{x}^0$.

Direct iteration. The three relaxed states are the end states of the leading blocks of (81) from $\boldsymbol{x}^0$:

1. mechanical: the state with $\hat{F}_M = 0$ on $\boldsymbol{x}^0 + \operatorname{span}\{\boldsymbol{d}_M\}$ with $p_I = p^*$, its own end pressure (End state of the mechanical step), the maximum of S on that path. Its path depends on the unknown p^*, so it is not the root of the leading 1×1 block with $\boldsymbol{D}$ held at $\boldsymbol{x}^0$; it is computed by the Newton iteration (131) below, or for a NASG liquid by the closed form below. With p_I held at $\boldsymbol{x}^0$ instead, the mechanical stage is the root of the leading 1×1 block with $\boldsymbol{D}$ held at $\boldsymbol{x}^0$;

2. thermo-mechanical: the leading 2×2 block, (85) with $\delta\tilde{m}_l^n = 0$ followed by (82), until $\hat{F}_M = \hat{F}_T = 0$. The end state is the maximum of S at fixed masses, the pT-relaxed state; for a NASG liquid it is computed by the closed form below instead;

3. full: the system (81)–(86), until $\hat{\boldsymbol{F}} = 0$. The end state is the maximum of S over all exchanges, $\boldsymbol{F} = 0$, the pTg-relaxed state.

The pT- and pTg-relaxed states are unique, so they do not depend on p_I or on the path. The unknowns of the iteration are $\tilde{V}_l$, $\tilde{E}_l$ and $u = \ln m_v$ (Dilute vapor), and the step is damped: for $\lambda = 1, \frac{1}{2}, \frac{1}{4}, \ldots$ the trial step has the mass $m_v^n(e^{\lambda\delta u^n} - 1)$, with δu^n from (86), and $\delta\tilde{E}_l$, $\delta\tilde{V}_l$ from (85) and (82) with $\lambda\hat{\boldsymbol{F}}^n$ in place of $\hat{\boldsymbol{F}}^n$ and this mass. The first λ for which both phases have a state of their equations of state ($V_k > 0$, $m_l > 0$, $m_g > 0$, ρ_l below the largest density of its equation of state ($1/b$ for NASG), $p_l + p_\infty > 0$, $p_g > 0$, $T_k > 0$, and $m_v > 0$ in the full system) is taken. At $\lambda = 1$ the trial is the Newton step, so near the end state the iteration converges quadratically; as $\lambda \to 0$ it tends to λ times the Newton step, and since the halving acts on the unknowns, a mass step δu of any size is reduced to an admissible one.

Pivots and convergence. The elimination divides by the pivots $\hat{J}_{MM}^n = -\mathcal{K}^{M,n}$, $\hat{J}_{TT}^{\prime n} = -\mathcal{K}^{T,n}$ and $\hat{J}_{CC}^{\prime\prime n} = -\mathcal{K}^{C,n}$ of the block. They are positive near the end state by (100); an iterate at which one of them is not positive stops the relaxation, and the cell is returned unchanged. The iteration

ends when each force of the block satisfies

$$|\hat{F}_k| \le \max\Big(\mathrm{tol}_k\, z_k^{\mathrm{ref}},\ 10\,\epsilon_{\mathrm{m}}\big[\,|z_k|+V\big(|Z_{l,k1}|+|Z_{g,k1}|\big)+\big(|E_l|+|E_g|\big)|Z_{l,k2}|+|E_g|\,|Z_{g,k2}|\,\big]\Big), \qquad k\in\{M,T,C\}, \tag{128}$$

with the tolerances tol_k, the scales $z^{\mathrm{ref}} = [\max(|p_l|,|p_g+p_j|),\ \max(T_l,T_g),\ R_vT_g]$, the magnitudes $|z_k| = |p_l|,\ \max(T_l,T_g),\ \max(|g_l|,|g_v|)$, $\boldsymbol{Z}_k \equiv \partial \boldsymbol{z}_k/\partial \boldsymbol{x}_k$ with the columns of (108), $V = V_l + V_g$ and the machine precision ϵ_{m}. The second term is the round-off of $\hat{F}_k$: $V_g = V - V_l$ and E_l are obtained from the conserved totals and carry their round-off, which $\boldsymbol{Z}_k$ amplifies.

Chemical stage on the pT-relaxed states. Started far from the pT-relaxed state, the mass step of (86) is driven by the faster forces through $\hat{F}_C''$, and a large mass step leaves the phases far from the pT-relaxed state through the linear $\delta\tilde{V}_l$ and $\delta\tilde{E}_l$ that accompany it. The pTg-relaxed state is therefore computed from $\boldsymbol{x}^0$ with the masses as the only outer unknown. With the air mass fixed, the masses are set by m_v. Let $S^*(m_v)$ be the maximum of S at fixed masses, attained at the unique pT-relaxed state. S is concave in the extensive state, and at fixed masses the maximization is over a convex set of (V_l, E_l), so S^* is concave in m_v. Along the pT-relaxed states $\hat{F}_M = \hat{F}_T = 0$, so by the end state of the chemical step $dS^* = \hat{F}_C\, dm_v/T$ with $dm_v = d\tilde{m}_l$, and $\hat{F}_C/T$ is non-increasing in m_v: it has at most one root, and its sign tells on which side of the root m_v lies. The iteration is:

1. the pT-relaxed state at the masses of $\boldsymbol{x}^0$ (item 2 above, or the closed form below for a NASG liquid), which gives the start m_v^0 on the pT-relaxed states;

2. at the iterate m_v^n, with $\boldsymbol{D}$ evaluated there ($p_I = p_l$, since the iterate is a pT-relaxed state, and v_l, e_l of the iterate) instead of at $\boldsymbol{x}^0$, the Newton step $\delta u^n = \delta\tilde{m}_l^n/m_v^n$ of (86); its pivot $-\mathcal{K}^C$ is the derivative of $\hat{F}_C$ along the pT-relaxed states by (98), so this is Newton's method in u on $\hat{F}_C(m_v)$. u^n is a lower bound of the root if $\hat{F}_C''^n > 0$ and an upper bound if $\hat{F}_C''^n < 0$. $\hat{F}_C''$ is used rather than $\hat{F}_C$ because the iterates hold $\hat{F}_M$ and $\hat{F}_T$ only to the tolerance of the iteration, and $\hat{F}_C''$ is the chemical force with these residuals eliminated, the one that sets the sign of δu^n. u^n is always one of the bounds, and a Newton point outside the bounds found so far is replaced by their midpoint;

3. the trial transfer: the mass $m_v^n(e^{\delta u}-1)$ with $\delta\tilde{E}_l$ and $\delta\tilde{V}_l$ from (85) and (82), which hold $\hat{F}_M$ and $\hat{F}_T$ to first order, followed by the pT-relaxed state at the new masses (item 2 above, or the closed form below for a NASG liquid). If that state does not exist, δu is halved;

until $\hat{\boldsymbol{F}} = 0$. Every accepted iterate is a pT-relaxed state, the bounds close on the unique root, and Newton's method converges quadratically near it.

Existence of the root. If the gas contains air, $m_a > 0$ keeps $V_g = (m_vR_v + m_aR_a)T/p_g$ positive as $m_v \to 0$, so $p_v \to 0$, $g_v \to -\infty$ and $\hat{F}_C \to +\infty$: a root exists. For a pure vapor, $m_a = 0$, the gas can vanish as $m_v \to 0$ and a root need not exist; the maximum of S is then on the boundary $m_g = 0$, and the cell condenses completely. As $m_v \to 0$ on the pT-relaxed states, $p_g \to 0$ and $\hat{F}_C \to +\infty$ unless the liquid holding all the water fills the volume at $p_l > p_j$; for a NASG liquid at temperature T, $p_l + p_\infty = R_lT/(v_l - b)$ with $R_l = (\gamma_l - 1)c_{v,l}$, so this requires the water density $(m_l+m_v)/V \ge 1/\big(b + R_lT/(p_\infty + p_j)\big)$. At $\alpha_l = \frac{1}{2}$ the water density is at most $(\rho_l+\rho_v)/2$, with $\rho_l < 1/b$, and below the critical point ($T < T_c$, $p_v < p_c$) it stays below this bound: complete condensation occurs only past the critical point, where the model is not applied, and is not treated.

Order. The p- and pT-relaxed states are computed by items 1 and 2 of the direct iteration, or for a NASG liquid by the closed forms below. For the pTg-relaxed state, a cell with $\alpha_l < \epsilon_n$ or $\alpha_l > 1 - \epsilon_n$ is first seeded by nucleation (Sec. 1.1.6); then the pT-relaxed state at the masses of $\boldsymbol{x}^0$ is computed, and the chemical stage on the pT-relaxed states follows. A cell with $\alpha_l < \epsilon_n$ or

$\alpha_l > 1-\epsilon_n$ that is not seeded is returned unchanged: a single-phase cell is then the maximum of S by (162), and a two-phase one is left to the caller. The direct iteration of the full system solves the same $\hat{\boldsymbol{F}} = 0$ and ends at the same state, but its mass step can leave the admissible states, so it is not used for the pTg-relaxed state; the chemical stage on the pT-relaxed states converges from any start for which the root exists.

Mechanical stage with $p_I = p^*$ for a general equation of state. With $p_I = p^*$ held, (102) gives the path at fixed masses in the volume $\delta \equiv V_l - V_l^0$ received by the liquid,

$$V_l = V_l^0 + \delta, \qquad E_l = E_l^0 - p^*\delta, \qquad V_g = V_g^0 - \delta, \qquad E_g = E_g^0 + (p^* - p_j)\,\delta, \tag{129}$$

which conserves V, the masses and $E_l + E_g + p_j V_l$ for any δ and p^*. The end state of the stage is the pressure equilibrium of the path whose interface pressure is its own end pressure, two conditions for the two unknowns δ and p^*:

$$R_l \equiv p_l(V_l, E_l) - p^* = 0, \qquad R_g \equiv p_g(V_g, E_g) + p_j - p^* = 0, \tag{130}$$

whose difference is $R_l - R_g = \hat{F}_M$. Along (129) at fixed p^* the pressures change by $\partial_M p_k$ of (111) with $p_I = p^*$, and at fixed δ a change of p^* moves the energies by $\mp\delta\, dp^*$, which changes p_k by $\mp\delta\,\Gamma_k/V_k$ by (108). Newton's method on (130) is therefore

$$\begin{bmatrix} \partial_M p_l & -1 - \delta\,\Gamma_l/V_l \\ \partial_M p_g & -1 + \delta\,\Gamma_g/V_g \end{bmatrix} \begin{bmatrix} \Delta\delta \\ \Delta p^* \end{bmatrix} = -\begin{bmatrix} R_l \\ R_g \end{bmatrix}, \tag{131}$$

with every coefficient at the current iterate. At $\delta = 0$ the determinant is $\partial_M p_g - \partial_M p_l = \mathcal{K}^M$ of (115) at $p_I = p^*$, positive by the End state of the mechanical step. Each iterate is evaluated on the path (129) from $\boldsymbol{x}^0$, not by increments, so the conservation holds exactly at every iterate; the iteration starts at $\delta = 0$ with p^* the impedance-weighted p_I of the Algorithm, step 2, is damped by halving the step until both phases have a state, as in the iteration of the stages, and ends when $\hat{F}_M$ and R_l are below the tolerance of $\hat{F}_M$ in (128). For a NASG liquid and the ideal-gas mixture its root is the closed form below.

Closed form of the mechanical stage. For a NASG liquid, $(p_l + \gamma_l p_\infty)(V_l - b m_l) = (\gamma_l - 1)(E_l - m_l q_l)$ ($b = 0$ for a stiffened gas), and for the ideal-gas mixture of fixed composition, $p_g V_g = \Gamma_g(E_g - Q_g)$ with $Q_g = m_v q_v + m_a q_a$. Let $\delta \equiv V_l - V_l^0 = V_g^0 - V_g$ be the volume received by the liquid, $W^0 \equiv V_l^0 - b m_l$, and $y \equiv p^* - p_j$ the gas pressure of the end state. By (102) with $p_I = p^*$, $E_l = E_l^0 - p^*\delta$ and $E_g = E_g^0 + y\,\delta$. The gas relation at the end state, $y(V_g^0 - \delta) = \Gamma_g(E_g^0 - Q_g) + \Gamma_g y\,\delta = p_g^0 V_g^0 + \Gamma_g y\,\delta$, gives

$$\delta = \frac{V_g^0\,(y - p_g^0)}{(1+\Gamma_g)\,y}, \tag{132}$$

which increases with y. The liquid relation, $(p^* + \gamma_l p_\infty)(W^0 + \delta) = (\gamma_l - 1)(E_l^0 - m_l q_l) - (\gamma_l - 1)p^*\delta = (p_l^0 + \gamma_l p_\infty)W^0 - (\gamma_l - 1)p^*\delta$, gives

$$\delta = \frac{W^0\,(p_l^0 - p^*)}{\gamma_l\,(p^* + p_\infty)}, \tag{133}$$

which decreases with $p^* = y + p_j$. With $z \equiv y - p_g^0$, so that $p_l^0 - p^* = \hat{F}_M^0 - z$, equating (132) and (133) and multiplying by $(1+\Gamma_g)\,y\,\gamma_l(p^* + p_\infty)$ gives

$$\begin{gathered} a\,z^2 + b\,z + c = 0, \qquad a = \gamma_l V_g^0 + (1+\Gamma_g)W^0, \\ b = a\,p_g^0 + \gamma_l V_g^0(p_j + p_\infty) - (1+\Gamma_g)W^0\hat{F}_M^0, \qquad c = -(1+\Gamma_g)W^0 p_g^0 \hat{F}_M^0. \end{gathered} \tag{134}$$

$a > 0$, and at $y = 0$, $z = -p_g^0$, the left side is $-\gamma_l V_g^0 p_g^0 (p_j + p_\infty) < 0$ for $p_j > -p_\infty$, so (134) has exactly one root with $y > 0$, its larger root; since (132) increases and (133) decreases, it is the only end state. It is computed without cancellation as $z = -2c/(b + \sqrt{b^2 - 4ac})$ for $b > 0$ and $z = (-b + \sqrt{b^2 - 4ac})/(2a)$ otherwise, δ follows from (132), and E_l, E_g from the path, so that (158) holds exactly. The end state is admissible: $y > 0$, $p^* + p_\infty > 0$, $V_g = V_g^0 (p_g^0 + \Gamma_g y)/\big((1 + \Gamma_g) y\big) > 0$ and $V_l - b m_l = W^0 \big(p_l^0 + \gamma_l p_\infty + (\gamma_l - 1) p^*\big)/\big(\gamma_l (p^* + p_\infty)\big) > 0$, so both temperatures are positive.

The NASG liquid is admissible for $V_l > b m_l$, where its pressure is finite; as $V_l \to b m_l$, $p_l \to \infty$, so the co-volume bounds the compression by itself and a relaxation from an admissible state stays admissible.

Closed form of the pT-relaxed state. At fixed masses the pT-relaxed state has $T_l = T_g = T$ and $p_l = p_g + p_j = y + p_j$, with y the gas pressure, and keeps the cell volume $V = V_l^0 + V_g^0$ and $E_l + E_g + p_j V_l$. With the NASG liquid and the ideal-gas mixture of the previous paragraph, write $W \equiv V_l - b m_l$ for the free volume of the liquid, $\omega \equiv V - b m_l$, $A_l \equiv m_l R_l$ with $R_l = (\gamma_l - 1) c_{v,l}$, $A_g \equiv m_v R_v + m_a R_a$, $C \equiv m_l c_{v,l} + C_{v,g}$ and $\pi \equiv p_\infty + p_j$. The liquid energy is $E_l = m_l (c_{v,l} T + q_l) + p_\infty W$ and the gas energy $E_g = C_{v,g} T + Q_g$, so the conservation of $E_l + E_g + p_j V_l$, with $V_l = W + b m_l$, reads

$$C T + \pi W = E_s, \qquad E_s \equiv E_l^0 + E_g^0 - m_l q_l - Q_g + p_j W^0. \tag{135}$$

The equations of state give $W = A_l T/(y + \pi)$, since $p_l + p_\infty = y + \pi$, and $V_g = A_g T/y$, and the volume is

$$T\Big(\frac{A_l}{y + \pi} + \frac{A_g}{y}\Big) = \omega. \tag{136}$$

Eliminating T between (135), $T\big(C + \pi A_l/(y + \pi)\big) = E_s$, and (136), and multiplying by $y(y + \pi)$,

$$a\, y^2 + b\, y + c = 0, \qquad a = C\omega, \qquad b = \omega\pi(C + A_l) - E_s(A_l + A_g), \qquad c = -E_s A_g \pi. \tag{137}$$

For an admissible start, $E_l^0 - m_l q_l = m_l c_{v,l} T_l^0 + p_\infty W^0$ and $E_g^0 - Q_g = C_{v,g} T_g^0$, so $E_s = m_l c_{v,l} T_l^0 + C_{v,g} T_g^0 + \pi W^0 > 0$ for $p_j > -p_\infty$; with $\omega > W^0 > 0$, $a > 0$ and $c < 0$, and (137) has exactly one root with $y > 0$, its larger root, computed without cancellation as $y = -2c/(b + \sqrt{b^2 - 4ac})$ for $b \geq 0$ and $y = (-b + \sqrt{b^2 - 4ac})/(2a)$ otherwise. This is the unique pT-relaxed state, consistently with its uniqueness as the maximum of S at fixed masses. It is admissible: $y > 0$, $p_l + p_\infty = y + \pi > 0$, and by (136) $T = \omega\, y(y + \pi)/\big(A_l y + A_g (y + \pi)\big) > 0$, so $W > 0$ and $V_g = \omega A_g (y + \pi)/\big(A_l y + A_g (y + \pi)\big) > 0$. The cell is updated with V_g, $E_g = C_{v,g} T + Q_g$ with $T = y V_g/A_g$, and E_l from the conservation of $E_l + E_g + p_j V_l$, which then holds exactly. The same conditions, $E_s > 0$, $\omega > 0$ and $p_j > -p_\infty$, decide whether the pT-relaxed state exists at the trial masses of the chemical stage on the pT-relaxed states.

Relation to Pelanti's sequential relaxation. Instantaneous relaxation needs only the state at which each force vanishes, not the size of the force. The entropy forces $\tilde{\boldsymbol{F}}$ and the forces $\hat{\boldsymbol{F}}$ are related by the invertible matrix $\boldsymbol{C}$ of (77), so they vanish at the same states and give the same end states. The steps above are moreover computed in $\hat{\boldsymbol{F}}$, whose components are Pelanti's forces [1]: $p_1 - p_2$, $T_1 - T_2$, $g_1 - g_2$ with phase 1 the liquid, $p_j = 0$, and g_2 the pure vapor. Her stiffnesses are those of the steps above:

- mechanical: $K_p = \mu\,(1/\xi_1^p + 1/\xi_2^p)$ [1, eq. (55)], where $1/\xi_k^p = [\Gamma_k (p_I - p_k) + \rho_k c_k^2]/\alpha_k$ [1, eq. (51)] is the phase-k term of $\mathcal{K}^M$ in (115), so $K_p = \mu\, \mathcal{K}^M$;
- thermal: per unit heat given by phase 1, $(\partial T_1/\partial \tilde{E}_l)_{\hat{F}_M, \tilde{m}_l} = -1/\xi_1^T$ and $(\partial T_2/\partial \tilde{E}_l)_{\hat{F}_M, \tilde{m}_l} = 1/\xi_2^T$ [1, eqs. (62), (63)], so $K_T = \vartheta\,(1/\xi_1^T + 1/\xi_2^T)$ [1, eq. (66)] is $\vartheta\, \mathcal{K}^T$ of (119);

- chemical: her S_p and S_T [1, eq. (23)] are the changes of p and T per unit mass received by phase 1 in the pT-relaxed model, $-(\partial p/\partial \tilde{m}_l)_{\hat{F}_M,\hat{F}_T}$ and $-(\partial T/\partial \tilde{m}_l)_{\hat{F}_M,\hat{F}_T}$, i.e. Π and Θ. By [1, eqs. (71b), (71c), (73)], $1/\xi_1^g = S_p/\rho_1 - s_1 S_T$ and $1/\xi_2^g = -S_p/\rho_2 + s_2 S_T$, so $K_g = \nu\,(1/\xi_1^g + 1/\xi_2^g) = \nu\,(\Delta v\,\Pi - \Delta s\,\Theta)$, which is $\nu\,\mathcal{K}^C$ with $\mathcal{K}_{CC} = 0$ for a pure vapor. Her eq. (72) prints $-s_2 S_T$ for $k = 2$; the sign used here follows from her eq. (71c).

Her instantaneous limits [1, eqs. (57), (68), (79), (80)] are therefore the transfers (101), with the heat and mass signs of footnote 1. She takes the volume fraction and the partial density from these transfers and then closes the pressure with the mixture energy relation [1, eqs. (59), (70), (82)], which is option 3 of the update of pressure and temperature, in one pass; the steps above as written use option 1. One pass of the steps (101) with option 3 is therefore her instantaneous procedure, step by step. Her one-pass states are first-order approximations of the exact relaxed states, which the iteration of the stages reaches: the pT- and pTg-relaxed states are unique, and the mechanical end state is the one of the path $\boldsymbol{d}_M$ with the chosen p_I.

Relation to Pelanti and Shyue's instantaneous relaxation. [2] relax instantaneously, $\mu, \theta, \nu \to \infty$ [2, eqs. (34), (41), (44)], and solve for the end states exactly: pressure equilibrium at fixed masses with the interface work $E_k^* - E_k^0 = \mp p_I(\alpha_1^* - \alpha_1^0)$, $p_I = (p_I^0 + p^*)/2$ [2, eqs. (37), (38)]; pressure and temperature equilibrium at fixed masses [2, eqs. (42), (43)]; pressure, temperature and Gibbs-energy equilibrium [2, eqs. (46), (47)]. The last two are the unique maxima of S at fixed masses and over all exchanges, i.e. the end states of the thermal and chemical stages, for any p_I. Their mechanical end state moves $\boldsymbol{x}_l$ along the single direction $\boldsymbol{d}_M$ built on the constant $p_I = (p_I^0 + p^*)/2$, with p_I^0 their impedance-weighted interface pressure, so it is the end state of the mechanical stage with that p_I, a value set implicitly by the end state. At their end state $\tilde{F}_M$ is half of (117) with $p_I = p_I^0$; the mechanical stage here takes $p_I = p^*$. For a stiffened-gas liquid ($b = 0$), $p_j = 0$ and a single gas, the root of (137) is the admissible root of their quadratic for the pT-relaxed pressure [2, eqs. (C.3)–(C.5)]. Their thermo-chemical step needs neither ρ_I nor e_I [2, eq. (44)], consistently with the independence of the chemical end state from $\boldsymbol{d}_C$ shown above.

1.1.5 Finite-rate relaxation

Starting from equation (24), we have

$$\frac{dS}{dt} = \boldsymbol{F} \cdot \frac{d\boldsymbol{x}_l}{dt} \tag{138}$$

Entropy maximization fixes the end state of each step, but not how fast it is approached. The rate requires a constitutive law for the transfers, which is additional physical input. As in the sequential relaxation, the mechanical, thermal and chemical processes are assumed to act on separated time scales, in this order from the fastest to the slowest [1, Sec. 5].

Entropy production of each step. With $d\boldsymbol{x}_l/dt = \boldsymbol{D}\,\dot{\tilde{\boldsymbol{x}}}_l$, (138) becomes $dS/dt = \tilde{\boldsymbol{F}} \cdot \dot{\tilde{\boldsymbol{x}}}_l = \tilde{F}_M \dot{\tilde{V}}_l + \tilde{F}_T \dot{\tilde{E}}_l + \tilde{F}_C \dot{\tilde{m}}_l$ by the same steps as (49). During a step the faster forces are held at zero and the slower transfers are inactive, so only the term of the step remains. By (79),

$$\frac{dS}{dt} = \frac{\hat{F}_M}{T_M}\,\dot{\tilde{V}}_l, \qquad \frac{dS}{dt} = \frac{\hat{F}_T}{T_l T_g}\,\dot{\tilde{E}}_l, \qquad \frac{dS}{dt} = \frac{\hat{F}_C}{T}\,\dot{\tilde{m}}_l, \tag{139}$$

in the mechanical, thermal and chemical step, respectively. The second law, $dS/dt \geq 0$, therefore requires that the liquid expands while $\hat{F}_M > 0$, gives heat while $\hat{F}_T > 0$, and gives mass while $\hat{F}_C > 0$.

Rate law. The simplest law with these signs is linear in the force of the step, with a positive rate times the liquid-minus-gas difference,

$$\dot{\tilde{V}}_l = \lambda \hat{F}_M, \qquad \dot{\tilde{E}}_l = \vartheta \hat{F}_T, \qquad \dot{\tilde{m}}_l = \nu \hat{F}_C, \tag{140}$$

with rates $\lambda, \vartheta, \nu \geq 0$ that may depend on the state. Substituting (140) into (139),

$$\frac{dS}{dt} = \frac{\lambda \hat{F}_M^2}{T_M}, \qquad \frac{dS}{dt} = \frac{\vartheta \hat{F}_T^2}{T_l T_g}, \qquad \frac{dS}{dt} = \frac{\nu \hat{F}_C^2}{T}, \tag{141}$$

all non-negative. With $\hat{\boldsymbol{F}}$ of (67), (140) reads

$$\dot{\tilde{\boldsymbol{x}}}_l = \boldsymbol{R}\,\hat{\boldsymbol{F}}, \qquad \boldsymbol{R} = \operatorname{diag}(\lambda, \vartheta, \nu), \tag{142}$$

each step using its own row. These are Pelanti's relaxation sources [1, eq. (2)] with phase 1 the liquid, up to the sign convention for heat and mass[1], and her mechanical rate μ is renamed λ to avoid a clash with the chemical potential. Departures from her model: p_j in $\hat{F}_M$, and $g_v(p_v, T_g)$ in place of her pure-vapor g_2. Written in the entropy forces instead, by (79), (140) is $\dot{\tilde{\boldsymbol{x}}}_l = \boldsymbol{L}\,\tilde{\boldsymbol{F}}$ with $\boldsymbol{L} = \operatorname{diag}(\lambda T_M,\, \vartheta T_l T_g,\, \nu T)$: the temperatures move into the coefficients. Pelanti allows her rates to be functions of the state [1, after eq. (2)], so both forms describe the same models; $\hat{\boldsymbol{F}}$ is used here so that the rates are hers.

Relaxation equation of each step. Along a step, with the faster forces held at zero, the force changes with the transferred amount through the stiffness (98). By the chain rule and (140),

$$\begin{aligned}
\frac{d\hat{F}_M}{dt} &= \left(\frac{\partial \hat{F}_M}{\partial \tilde{V}_l}\right)_{\tilde{E}_l, \tilde{m}_l} \dot{\tilde{V}}_l = -\mathcal{K}^M \lambda \hat{F}_M, \\
\frac{d\hat{F}_T}{dt} &= \left(\frac{\partial \hat{F}_T}{\partial \tilde{E}_l}\right)_{\hat{F}_M, \tilde{m}_l} \dot{\tilde{E}}_l = -\mathcal{K}^T \vartheta \hat{F}_T, \\
\frac{d\hat{F}_C}{dt} &= \left(\frac{\partial \hat{F}_C}{\partial \tilde{m}_l}\right)_{\hat{F}_M, \hat{F}_T} \dot{\tilde{m}}_l = -\mathcal{K}^C \nu \hat{F}_C.
\end{aligned} \tag{143}$$

In vector form, (98) reads $d\hat{\boldsymbol{F}} = -\boldsymbol{\mathcal{K}}\, d\tilde{\boldsymbol{x}}_l$ row by row, with $\boldsymbol{\mathcal{K}} \equiv \operatorname{diag}(\mathcal{K}^M, \mathcal{K}^T, \mathcal{K}^C)$, and with (142), since diagonal matrices commute,

$$\frac{d\hat{\boldsymbol{F}}}{dt} = -\boldsymbol{\mathcal{K}}\boldsymbol{R}\,\hat{\boldsymbol{F}} = -\boldsymbol{R}\boldsymbol{\mathcal{K}}\,\hat{\boldsymbol{F}}, \tag{144}$$

each step using its own row.

[1] Pelanti [1, eq. (2)] counts all transfers as received by phase 1, so her heat and mass sources are $\vartheta(T_2 - T_1)$ and $\nu(g_2 - g_1)$, a positive rate times gas minus liquid. Here heat and mass are counted as given by the liquid, so her sources are $-\dot{\tilde{E}}_l = -\vartheta \hat{F}_T$ and $-\dot{\tilde{m}}_l = -\nu \hat{F}_C$ with the same rates; her directions are $-\boldsymbol{d}_T$ and $-\boldsymbol{d}_C$, and her transfers are $-\delta \tilde{E}_l$ and $-\delta \tilde{m}_l$.

Solution with frozen coefficients. Freezing $\boldsymbol{R}$ and $\boldsymbol{\mathcal{K}}$ at the start state of the step, where the force is $\hat{\boldsymbol{F}}^0$, (144) has the exact solution

$$\hat{\boldsymbol{F}}(t) = e^{-\boldsymbol{R}\boldsymbol{\mathcal{K}}t}\,\hat{\boldsymbol{F}}^0, \tag{145}$$

where the exponential of the diagonal matrix is taken entry by entry. Substituting (145) into (140) and integrating over $[0, \Delta t]$, for the mechanical step

$$\delta\tilde{V}_l = \int_0^{\Delta t} \lambda \hat{F}_M^0\, e^{-\lambda\mathcal{K}^M t}\, dt = \frac{\hat{F}_M^0}{\mathcal{K}^M}\big(1 - e^{-\lambda\mathcal{K}^M\Delta t}\big), \tag{146}$$

and likewise for the thermal and chemical steps,

$$\delta\tilde{\boldsymbol{x}}_l = \boldsymbol{\Omega}\,\delta\tilde{\boldsymbol{x}}_l^{\infty}, \qquad \boldsymbol{\Omega} \equiv \boldsymbol{I} - e^{-\boldsymbol{R}\boldsymbol{\mathcal{K}}\Delta t} = \mathrm{diag}(\omega_M, \omega_T, \omega_C), \qquad \delta\tilde{\boldsymbol{x}}_l^{\infty} = \boldsymbol{\mathcal{K}}^{-1}\hat{\boldsymbol{F}}^0, \tag{147}$$

$\delta\tilde{\boldsymbol{x}}_l^{\infty} = [\hat{F}_M^0/\mathcal{K}^M,\, \hat{F}_T^0/\mathcal{K}^T,\, \hat{F}_C^0/\mathcal{K}^C]^T$ holds the instantaneous transfers (101) of the sequential relaxation. Every increment of a sequential step (δp, δT, δV, δE, δm) is linear in its transfer, so a finite-rate step is the instantaneous step with all of its increments multiplied by the entry of $\boldsymbol{\Omega}$ of the step, $\omega \in [0, 1)$. $\omega \to 1$ as the rate goes to infinity, which recovers the instantaneous sequential relaxation, i.e. the Newton step of the constrained entropy maximum of the step; $\omega = 0$ for zero rate. Within the linearization the force keeps its sign and decays monotonically, so by (141) the entropy increases throughout the step and the state does not pass the end state of the step. The entries of $\boldsymbol{R}\boldsymbol{\mathcal{K}}$, $\lambda\mathcal{K}^M$, $\vartheta\mathcal{K}^T$, $\nu\mathcal{K}^C$, are Pelanti's exponents K_p, K_T, K_g [1, eqs. (55), (66), (76)], and (147) is her solution [1, eqs. (54), (56), (65), (67), (75), (77)], up to the closure noted in the relation to Pelanti's sequential relaxation.

Chemical step with exact vapor potential. The frozen solution (147) treats $\mathcal{K}^C$ as constant over the step. For the ideal-gas mixture, $p_g = (m_v R_v + m_a R_a)T_g/V_g$ and the definition of x_v before (110) give the vapor partial pressure

$$p_v = x_v p_g = \frac{m_v R_v T_g}{V_g}, \tag{148}$$

and $g_v(p_v, T_g)$ contains $R_v T_g \ln p_v$, so at fixed V_g and T_g it changes by $R_v T_g/m_v$ per unit vapor mass, for a vapor–air gas and for a pure vapor alike. $\mathcal{K}^C$ therefore changes by $\mathcal{O}(1)$ relative to itself when m_v changes by $\mathcal{O}(m_v)$, and tends to infinity as $m_v \to 0$. Two failures of the frozen solution follow. Since $\hat{F}_C$ contains $-R_v T \ln m_v$, $\hat{F}_C^{**}/\mathcal{K}^C \sim m_v \ln(1/m_v) \to 0$ as $m_v \to 0$: it transfers almost nothing into a nearly dry gas, while the rate law (140) evaporates at the finite rate $\nu\hat{F}_C$. When a vapor-rich gas condenses, $\mathcal{K}^C$ grows as m_v falls, the frozen $\mathcal{K}^C$ underestimates it, and the frozen solution can condense more vapor than the gas holds. The step is therefore integrated along its linear path, with Π, Θ and the volume transfer per unit mass frozen at the start state $**$, as Pelanti freezes her capacities, and with g_l and g_v evaluated from the equations of state on this path.

Write $\tilde{m} \equiv m_v - m_v^{**}$ for the mass given by the liquid during the step, and $\chi_V \equiv \delta\tilde{V}_l/\delta\tilde{m}_l$ for the volume transfer per unit mass of (124). By (125) and (102), $\delta V_l = (\chi_V - v_l^{**})\,\delta\tilde{m}_l$, and the path of the step is

$$p_l = p_l^{**} - \Pi\,\tilde{m}, \qquad T = T^{**} - \Theta\,\tilde{m}, \qquad V_g = V_g^{**} - (\chi_V - v_l^{**})\,\tilde{m}. \tag{149}$$

On this path the force is

$$\hat{F}_C(m_v) = g_l(p_l, T) - g_v(p_v, T), \qquad p_v = \frac{m_v R_v T}{V_g}. \tag{150}$$

By (4), $dg_l = v_l\,dp_l - s_l\,dT$ and $dg_v = v_v\,dp_v - s_v\,dT$ with $v_v\,dp_v = R_vT\,d\ln p_v$, and by (148) $d\ln p_v = dm_v/m_v + dT/T - dV_g/V_g$. Along (149), $d\tilde{m} = dm_v$, so

$$\mathcal{K}^C(m_v) \equiv -\frac{d\hat{F}_C}{dm_v} = v_l\Pi - s_l\Theta + R_vT\Big(\frac{1}{m_v} - \frac{\Theta}{T} + \frac{\chi_V - v_l^{**}}{V_g}\Big) + s_v\Theta, \tag{151}$$

with v_l, s_l at (p_l, T) and s_v at (p_v, T). At m_v^{**} this is $\mathcal{K}^C$ of (127): there p_g changes as p_l, since $\hat{F}_M$ is held, so differentiating $p_g = (m_vR_v + m_aR_a)T/V_g$ along the path gives $-\Pi = R_vT/V_g - p_g\Theta/T + p_g(\chi_V - v_l)/V_g$. Multiplying by $\bar{v}_v = R_vT/p_g$, with $R_vT\,\bar{v}_v/V_g = R_vTx_v/m_v$, gives $-\bar{v}_v\Pi = R_vTx_v/m_v - R_v\Theta + R_vT(\chi_V - v_l)/V_g$, and (151) becomes $(v_l - \bar{v}_v)\Pi - (s_l - \bar{s}_v)\Theta + R_vT(1-x_v)/m_v = \Delta v\,\Pi - \Delta s\,\Theta + \mathcal{K}_{CC}$. With $\hat{F}_C$ a function of m_v alone, the rate law $dm_v/dt = \nu\hat{F}_C(m_v)$ is a separable scalar equation, and the vapor mass at the end of the step solves

$$\nu\,\Delta t = \int_{m_v^{**}}^{m_v^{\otimes}} \frac{dm_v}{\hat{F}_C(m_v)}, \qquad \delta\tilde{m}_l = m_v^{\otimes} - m_v^{**}, \tag{152}$$

after which $\delta\tilde{E}_l$ and $\delta\tilde{V}_l$ follow from (124). Where $\mathcal{K}^C(m_v) \equiv -d\hat{F}_C/dm_v > 0$, $\hat{F}_C$ decreases monotonically and has a single root m_v^{∞}, the end state of the step with these frozen coefficients. Near the root $\hat{F}_C \approx \mathcal{K}^C(m_v^{\infty})(m_v^{\infty} - m_v)$, so the integral diverges logarithmically there: $m_v^{\otimes}$ lies between m_v^{**} and m_v^{∞}, tends to m_v^{∞} as $\nu\Delta t \to \infty$, and, where the path is admissible down to $m_v \to 0$, m_v stays positive because $\hat{F}_C \to +\infty$ as $m_v \to 0$. Two limits recover the known solutions. Linearizing $\hat{F}_C$ about m_v^{**} gives $\hat{F}_C = \hat{F}_C^{**} - \mathcal{K}^C\tilde{m}$ and (147); for a pure vapor this is Pelanti's solution, which therefore agrees with (152) to first order in the transferred mass. For a dilute vapor in a gas whose pressure, temperature and volume do not change with the transfer ($\Pi = \Theta = 0$, $\chi_V = v_l^{**}$), $\hat{F}_C = R_vT\ln(m_s/m_v)$ with $m_s = m_v^{**}e^{\hat{F}_C^{**}/(R_vT)}$ the vapor mass at saturation, and (152) is an exponential integral.

The integral is evaluated in a variable θ that removes its end-point singularities: $m_v = m_v^{\infty}/(1 + e^{-\theta})$ for evaporation and $m_v = m_v^{\infty} + (m_v^{**} - m_v^{\infty})e^{-\theta}$ for condensation. The integrand $|dm_v/d\theta|/|\hat{F}_C|$ is then bounded, vanishes as $m_v \to 0$ and tends to $1/\mathcal{K}^C(m_v^{\infty})$ at the root. Gauss–Legendre quadrature on unit panels in θ accumulates $\nu t(\theta)$, and the upper limit of (152) is found by Newton's method within the panel where νt reaches $\nu\Delta t$. The root m_v^{∞} is bracketed from m_v^{**} by the sign of $\hat{F}_C$, symmetrically for the two directions: for evaporation ($\hat{F}_C^{**} > 0$) m_v is doubled, starting from $\max(2m_v^{**}, m_v^d)$ with m_v^d the vapor mass of mole fraction 10^{-12}, until $\hat{F}_C < 0$; for condensation ($\hat{F}_C^{**} < 0$) it is halved until $\hat{F}_C > 0$. Each trial must lie on the admissible part of the path (149), $V_g > 0$ with a liquid state at (p_l, T), and leave mass in the phase that gives it, $m_v < m_v^{**} + m_l^{**}$ for evaporation and $m_v > 0$ for condensation; otherwise the path has no root and the step fails. Within the bracket the root is found by Newton's method in $\ln m_v$, with bisection when a Newton point leaves the bracket.

The chemical rate may instead be set by a relaxation time τ_C, a rate that depends on the state as (140) allows: $\nu = 1/\big(\mathcal{K}^C(m_v^{\infty})\,\tau_C\big)$, with the stiffness (151) at the end state of the step. Near that state $\hat{F}_C \approx \mathcal{K}^C(m_v^{\infty})(m_v^{\infty} - m_v)$, so m_v approaches m_v^{∞} as e^{-t/τ_C} in every cell, whereas $1/(\nu\mathcal{K}^C)$ at a fixed ν varies between cells with $\mathcal{K}^C \sim R_vT/m_v$. The stiffness at the start state would not do: it diverges as $m_v^{**} \to 0$, so $\nu \to 0$ and a nearly dry gas would not evaporate.

Choice of the solution. The frozen solution (147), $\delta\tilde{m}_l = \omega_C\hat{F}_C^{**}/\mathcal{K}^C$ with $\mathcal{K}^C$ at $**$, is the linearization of (152), accurate for small $\nu\mathcal{K}^C\Delta t$, which is the regime the finite rate is meant for. It fails where $\mathcal{K}^C$ changes over the step. Its term $R_vT(1-x_v)/m_v$ changes relatively by $|\delta\tilde{m}_l|/m_v^{**}$, while its other terms change with p_l, T and V_g, whose relative changes along (149) are smaller for the same transfer. In condensation $\mathcal{K}^C$ grows as the vapor is removed, the frozen $\mathcal{K}^C$ underestimates

it, and the frozen transfer can exceed m_v^{**}; in evaporation into a nearly dry gas $\mathcal{K}^C$ falls as the vapor grows, the frozen $\mathcal{K}^C$ overestimates it, and the frozen transfer, about $m_v^{**}\ln(m_v^\infty/m_v^{**})$ for large $\nu\mathcal{K}^C\Delta t$, falls far short of the rate law. The chemical step therefore takes the frozen solution where $|\delta\tilde{m}_l| \le \epsilon_{\text{lin}}\, m_v^{**}$ and $m_l^{**} - \delta\tilde{m}_l > 0$, with $\epsilon_{\text{lin}} < 1$, so that $\mathcal{K}^C$ changes by at most of order ϵ_{lin} over the step and the relative error of the transfer is of order ϵ_{lin}; otherwise it takes (152) at the same ν. The test uses only the start state, so it is decided before the step. For a small step $|\delta\tilde{m}_l|/m_v^{**} \approx \nu\mathcal{K}^C\Delta t\,|\hat{F}_C^{**}|/(\mathcal{K}^C m_v^{**})$, the product of $\nu\mathcal{K}^C\Delta t$ and of the relative distance of m_v^{**} from its chemical equilibrium. A dry gas, $m_v^{**} = 0$, always takes (152): there $\hat{F}_C^{**}$ and $\mathcal{K}^C$ are infinite, the frozen solution is undefined, and from a seeded vapor mass it would transfer about $m_v^s\ln(m_v^\infty/m_v^s)$ whatever ν, so that the seed m_v^s, not the rate, would set the evaporation. With the relaxation time τ_C, ν requires m_v^∞, so the root is computed first and the test then decides whether the quadrature is needed. The integral is a safeguard for large steps and a dry gas; the rates of an application are meant to keep the chemical step in the frozen solution.

Activation. In one relaxation call the transfers faster than the finite-rate one are instantaneous ($r \to \infty$) and the slower ones are not activated, as in Pelanti's steps 2(a)–2(c) [1, Sec. 5], where the mechanical step is always instantaneous. After a linearized finite step the faster equilibria hold only to first order; they can be restored by iterating the faster stages with option 2 of the update of pressure and temperature.

Algorithm of a relaxation call. The relaxation acts on each cell after the hyperbolic step of size Δt; the momentum is not changed. A cell in which one phase is absent or nearly absent is first seeded by the nucleation of Sec. 1.1.6 when the chemical step is activated. Per unit cell volume, $V_l = \alpha_l$ and $V_g = \alpha_g = 1 - \alpha_l$, and the cell state is

$$\alpha_l, \qquad m_l = \alpha_l\rho_l, \qquad m_g = \alpha_g\rho_g, \qquad E_l = \alpha_l\rho_l e_l, \qquad E_g = \alpha_g\rho_g e_g, \qquad m_v = \alpha_g\rho_g Y_v, \quad (153)$$

with the air mass $m_a = m_g - m_v$ fixed. The steps are taken in the order M, T, C. With s the slowest activated step, the steps before s are instantaneous ($\omega = 1$), the step s has the rate $r_s \in \{\lambda, \vartheta, \nu\}$ ($\omega_s = 1$ if $r_s = \infty$), and the steps after s are skipped. Each step consists of the following parts, with every coefficient evaluated at the start state of the step and $\boldsymbol{D}$ held there.

1. Phase state. From the equations of state at the current cell state: ρ_k, e_k, p_k, T_k, the sound speed c_k with $K_{s,k} = \rho_k c_k^2$, Γ_k, and $C_{v,k} = m_k c_{v,k}$; for the liquid $v_l = 1/\rho_l$, h_l, s_l and $g_l(p_l, T_l)$; for the gas $x_v = p_v/p_g$, $\bar{v}_v = R_v T_g/p_g$, $\bar{s}_v = s_v(p_v, T_g)$, $g_v(p_v, T_g)$, $e_v = c_{v,v}T_g + q_v$ and $\mathcal{K}_{CC} = R_v T_g(1 - x_v)/m_v$. For the ideal-gas mixture $\Gamma_g = (m_v R_v + m_a R_a)/C_{v,g}$ and $K_{s,g} = (1 + \Gamma_g)\, p_g$.

2. Mechanical step. p_I is chosen between $p_g + p_j$ and p_l and held during the step. The default is Pelanti's impedance-weighted interface pressure [1], $p_I = \big(Z_g p_l + Z_l(p_g + p_j)\big)/(Z_l + Z_g)$ with $Z_k = \rho_k c_k$, for which $p_I - p_g - p_j = Z_g\hat{F}_M/(Z_l + Z_g)$ and $p_l - p_I = Z_l\hat{F}_M/(Z_l + Z_g)$. It keeps both numerators of $\mathcal{K}^M$ in (115) positive except for extreme forces. For the ideal-gas mixture, $K_{s,g} = \gamma_g p_g$ and $\Gamma_g = \gamma_g - 1$, so the gas numerator is $\gamma_g p_g + (\gamma_g - 1)Z_g\hat{F}_M/(Z_l + Z_g)$, negative only for $-\hat{F}_M > \gamma_g p_g(Z_l + Z_g)/\big((\gamma_g - 1)Z_g\big)$. For a stiffened-gas or NASG liquid, $K_{s,l} = \gamma_l(p_l + p_\infty)/(1 - b\rho_l)$ and $\Gamma_l = (\gamma_l - 1)/(1 - b\rho_l)$, so the liquid numerator is $\big(p_l + \gamma_l p_\infty + (\gamma_l - 1)p_I\big)/(1 - b\rho_l)$, positive for $p_I > -(p_l + \gamma_l p_\infty)/(\gamma_l - 1)$. A p_I close to the pressure of a liquid in tension, such as $p_I = p_l$ or the mean of p_l and $p_g + p_j$, can instead make $\mathcal{K}^M$ negative when the gas volume is small, as shown

after (100). With this p_I, by (115),

$$\hat{F}^0_M = p_l - p_g - p_j, \qquad \mathcal{K}^M = \frac{K_{s,l} + \Gamma_l(p_I - p_l)}{\alpha_l} + \frac{K_{s,g} + \Gamma_g(p_I - p_j - p_g)}{\alpha_g},$$
$$\omega_M = 1 - e^{-\lambda \mathcal{K}^M \Delta t}, \qquad \delta\tilde{V}_l = \omega_M \frac{\hat{F}^0_M}{\mathcal{K}^M}, \tag{154}$$
$$\alpha_l \leftarrow \alpha_l + \delta\tilde{V}_l, \qquad E_l \leftarrow E_l - p_I\,\delta\tilde{V}_l, \qquad E_g \leftarrow E_g + (p_I - p_j)\,\delta\tilde{V}_l,$$

with m_l, m_g, m_v unchanged. The linear increments of the intensive state are (116). An instantaneous mechanical step ($\lambda = \infty$) takes instead $p_I = p^*$, the pressure of its end state (End state of the mechanical step), by (131), or by (134) for a NASG liquid.

3. Thermal step. From a state with $\hat{F}_M = 0$, with p_I held at p_l, by (119),

$$\hat{F}^*_T = T_l - T_g, \qquad \mathcal{K}^T = \frac{1}{C_{v,l}} + \frac{1}{C_{v,g}} - \frac{\big(\Gamma_l T_l/\alpha_l + \Gamma_g T_g/\alpha_g\big)\big(\Gamma_l/\alpha_l + \Gamma_g/\alpha_g\big)}{K_{s,l}/\alpha_l + K_{s,g}/\alpha_g},$$
$$\omega_T = 1 - e^{-\vartheta \mathcal{K}^T \Delta t}, \qquad \delta\tilde{E}_l = \omega_T \frac{\hat{F}^*_T}{\mathcal{K}^T}, \qquad \delta\tilde{V}_l = -\frac{\Gamma_l/\alpha_l + \Gamma_g/\alpha_g}{K_{s,l}/\alpha_l + K_{s,g}/\alpha_g}\,\delta\tilde{E}_l, \tag{155}$$
$$\alpha_l \leftarrow \alpha_l + \delta\tilde{V}_l, \qquad E_l \leftarrow E_l - p_I\,\delta\tilde{V}_l - \delta\tilde{E}_l, \qquad E_g \leftarrow E_g + (p_I - p_j)\,\delta\tilde{V}_l + \delta\tilde{E}_l,$$

with m_l, m_g, m_v unchanged. The linear increments of the intensive state are (121).

4. Chemical step. From a state with $\hat{F}_M = \hat{F}_T = 0$, $T_l = T_g = T$, with p_I held at p_l: the entries (118) and (123) with $V_k = \alpha_k$, and $\mathcal{K}^M$, $\mathcal{K}^T$ of (154) and (155) at this state, give by (124), (125) and (127) the volume and heat per unit mass and the stiffness,

$$\chi_E \equiv \frac{\delta\tilde{E}_l}{\delta\tilde{m}_l} = \frac{1}{\mathcal{K}^T}\Big(\hat{J}_{TC} + \frac{\hat{J}_{TM}\hat{J}_{MC}}{\mathcal{K}^M}\Big), \qquad \chi_V \equiv \frac{\delta\tilde{V}_l}{\delta\tilde{m}_l} = \frac{\hat{J}_{MT}\,\chi_E + \hat{J}_{MC}}{\mathcal{K}^M},$$
$$\Pi = \frac{K_{s,l}}{\alpha_l}\,\chi_V + \frac{\Gamma_l}{\alpha_l}\,\chi_E, \qquad \Theta = \frac{\Gamma_l T}{\alpha_l}\,\chi_V + \frac{\chi_E}{C_{v,l}}, \qquad \mathcal{K}^C = (v_l - \bar{v}_v)\,\Pi - (s_l - \bar{s}_v)\,\Theta + \mathcal{K}_{CC}, \tag{156}$$

and the updates

$$\hat{F}^{**}_C = g_l - g_v, \qquad \omega_C = 1 - e^{-\nu \mathcal{K}^C \Delta t}, \qquad \delta\tilde{m}_l = \omega_C \frac{\hat{F}^{**}_C}{\mathcal{K}^C}, \qquad \delta\tilde{E}_l = \chi_E\,\delta\tilde{m}_l, \qquad \delta\tilde{V}_l = \chi_V\,\delta\tilde{m}_l,$$
$$\alpha_l \leftarrow \alpha_l + \delta\tilde{V}_l - v_l\,\delta\tilde{m}_l, \qquad m_l \leftarrow m_l - \delta\tilde{m}_l, \qquad m_g \leftarrow m_g + \delta\tilde{m}_l, \qquad m_v \leftarrow m_v + \delta\tilde{m}_l,$$
$$E_l \leftarrow E_l - p_I\,\delta\tilde{V}_l - \delta\tilde{E}_l - e_l\,\delta\tilde{m}_l, \qquad E_g \leftarrow E_g + (p_I - p_j)\,\delta\tilde{V}_l + \delta\tilde{E}_l + (e_l + p_j v_l)\,\delta\tilde{m}_l. \tag{157}$$

The linear increments of the intensive state are $\delta p^C = -\Pi\,\delta\tilde{m}_l$ and $\delta T^C = -\Theta\,\delta\tilde{m}_l$ for both phases. $\delta\tilde{m}_l$ is computed instead from (152), which evaluates g_l and g_v exactly along the path (149); $\omega_C \hat{F}^{**}_C/\mathcal{K}^C$ is its linearization.

5. Intensive state. After each step the intensive state is updated by one of the options of the update of pressure and temperature: the linear increments above (option 1), the equations of state at the updated cell state (option 2), or Pelanti's mixture-energy closure (option 3). When the steps up to s are instantaneous, the relaxed state is computed instead by the iteration of the stages with option 2, or for a NASG liquid by the closed forms of the mechanical stage and the pT-relaxed state; with a finite step s, the faster equilibria are reached by that iteration before the step, and may be restored the same way after it (Activation).

Conservation. The updates (154)–(157) follow (102), so for any p_I and any ω each step satisfies exactly

$$\delta\alpha_l + \delta\alpha_g = 0, \qquad \delta m_l + \delta m_g = 0, \qquad \delta m_a = 0, \qquad \delta E_l + \delta E_g + p_j\,\delta\alpha_l = 0. \tag{158}$$

Without surface tension ($p_j = 0$) and with Pelanti's sign convention (footnote 1), (154)–(157) are her finite-rate steps [1, eqs. (54)–(56), (65)–(67), (75)–(77)] with her stiffnesses, up to the closure of the intensive state.

Simultaneous transfers. Without rate separation the three transfers act at once, and $dS/dt = \tilde{F}_M \dot{\tilde{V}}_l + \tilde{F}_T \dot{\tilde{E}}_l + \tilde{F}_C \dot{\tilde{m}}_l$. $\tilde{F}_M = \hat{F}_M/T_M$ and $\tilde{F}_T = \hat{F}_T/(T_l T_g)$ hold at any state, but $\tilde{F}_C = \hat{F}_C/T$ only at pressure and temperature equilibrium; in general $\tilde{F}_C = (\hat{F}_C - v_l \hat{F}_M + s_l \hat{F}_T)/T_g$ by (77). Pelanti's model [1, eqs. (1e), (1f)] lets the transferred mass carry the energy g_I and no volume; for mass given by the liquid this is the direction $\boldsymbol{d}_C^P = [0,\, -g_I,\, -1]^T$, which is related to $\boldsymbol{d}_C$ by

$$\boldsymbol{d}_C = \boldsymbol{d}_C^P - v_l\, \boldsymbol{d}_M + (e_l + p_I v_l - g_I)\, \boldsymbol{d}_T. \tag{159}$$

Its conjugate force is $\tilde{F}_C^P = \boldsymbol{d}_C^P \cdot \boldsymbol{F} = -g_I F_2 - F_3 = (g_l - g_I)/T_l + (g_I - g_v)/T_g$. The interface chemical potential lies between g_l and g_v [1, eq. (3)]; with $g_l - g_I = w_C(g_l - g_v)$ and $g_I - g_v = (1 - w_C)(g_l - g_v)$, $w_C \in [0, 1]$, the same steps as for $\tilde{F}_M$ in (79) give, at any state,

$$\tilde{F}_C^P = \frac{w_C \hat{F}_C}{T_l} + \frac{(1 - w_C)\hat{F}_C}{T_g} = \frac{\hat{F}_C}{T_C}, \qquad \frac{1}{T_C} \equiv \frac{w_C}{T_l} + \frac{1 - w_C}{T_g}. \tag{160}$$

With $\boldsymbol{d}_C^P$, (142) therefore gives $dS/dt = \lambda \hat{F}_M^2/T_M + \vartheta \hat{F}_T^2/(T_l T_g) + \nu \hat{F}_C^2/T_C \geq 0$ for simultaneous transfers as well. Under rate separation the choice between $\boldsymbol{d}_C$ and $\boldsymbol{d}_C^P$ does not matter, since $\tilde{F}_M = \tilde{F}_T = 0$ there by (79).

Open: (i) solution of simultaneous transfers; (ii) values of λ, ϑ, ν.

1.1.6 Nucleation

The relaxation above needs both phases in the cell: the potentials $\boldsymbol{y}_k$ of (22) are intensive and undefined for a phase of zero mass. A cell that holds a single phase lies on the boundary $m_l = 0$ or $m_g = 0$ of the admissible states. It is the maximum of S over all exchanges only if no creation of a small amount of the absent phase increases S. Otherwise the phase present is metastable, a superheated liquid or a supersaturated gas: a seed of the absent phase is then created, and the steps M, T, C start from the two-phase state.

Entropy of a seed. Let P denote the phase present, the parent, and let the seed be a mass δm of the absent phase, a pure substance, at a uniform state (p_s, T_s) with specific volume v_s, energy e_s, entropy s_s and Gibbs energy $g_s = e_s + p_s v_s - T_s s_s$ by (2). The seed is exchanged under the conservation of the relaxation, $d\boldsymbol{x}_g = -\boldsymbol{A}\, d\boldsymbol{x}_l$ with $\boldsymbol{A}$ of (28) and the pressure jump p_j of the cell ($p_j = 0$ in a cell without interface). Let $\boldsymbol{\sigma} \equiv [v_s, e_s, 1]^T$ be the extensive state of the seed per unit mass. The parent changes by $\delta \boldsymbol{x}_P = -\delta m\, \boldsymbol{A}_P \boldsymbol{\sigma}$, with $\boldsymbol{A}_P = \boldsymbol{A}^{-1}$ for a liquid parent, whose seed is gas, and $\boldsymbol{A}_P = \boldsymbol{A}$ for a gas parent: per unit seed mass the liquid gives the energy $e_s - p_j v_s$ and the gas gives $e_s + p_j v_s$, so that $E_l + E_g + p_j V_l$ is conserved. The mass given is m_l for a liquid parent and m_v, at fixed m_a, for a gas parent. S is extensive, so the seed has the entropy $\delta m\, s_s$, and by (22) the entropy of the parent changes by $-\boldsymbol{y}_P \cdot \boldsymbol{A}_P \boldsymbol{\sigma}\, \delta m$ to first order. The entropy gained per unit seed mass is therefore

$$a \equiv \frac{dS}{d\delta m} = s_s - \boldsymbol{y}_P \cdot \boldsymbol{A}_P \boldsymbol{\sigma} = \frac{g_P - B_s}{T_P}, \qquad B_s \equiv e_s + p_P^* v_s - T_P s_s, \tag{161}$$

with $g_P = g_l(p_l, T_l)$ for the liquid by (16), $g_P = g_v(p_v, T_g)$ for the gas by (20), and $p_P^* = p_l - p_j$ for a liquid parent, $p_P^* = p_g + p_j$ for a gas parent: the pressure of the seed phase in Laplace equilibrium with the parent.

Best seed. At fixed (p_P^*, T_P), B_s is a function of the seed state. By (1), $dB_s = (T_s - T_P)\,ds_s - (p_s - p_P^*)\,dv_s$, so B_s is stationary at $T_s = T_P$, $p_s = p_P^*$. It is a minimum there, because $e_s(s_s, v_s)$ is convex for a stable phase and B_s differs from it by a linear function of (s_s, v_s); its minimum is $g_s(p_P^*, T_P)$. Hence some seed increases S to first order if and only if

$$g_P > g_s(p_P^*, T_P)\,: \tag{162}$$

the absent phase, at the temperature of the parent and in Laplace equilibrium with it, has the lower chemical potential. If (162) fails, the single-phase state is the maximum of S and the cell is not relaxed. For a seed at the temperature of the parent, $T_s = T_P = T$, (161) with $B_s = g_s + (p_P^* - p_s)v_s$ reads

$$a = \frac{g_P - g_s(p_s, T) + (p_s - p_P^*)\,v_s}{T}. \tag{163}$$

Cavitation. The parent is the liquid and the seed is pure vapor. Define the saturation pressure $p_{vsat}(p_l, T)$ of the liquid as the pressure of the vapor in chemical equilibrium with the liquid at (p_l, T),

$$g_v(p_{vsat}, T) = g_l(p_l, T). \tag{164}$$

By (4), $(\partial g_v/\partial p)_T = v_v > 0$, so $g_v(p_l - p_j, T_l) < g_l(p_l, T_l) = g_v(p_{vsat}, T_l)$ if and only if $p_l - p_j < p_{vsat}$, and (162) is

$$p_{vsat}(p_l, T_l) > p_l - p_j. \tag{165}$$

For the NASG liquid and the ideal-gas vapor, $g = c_p T + q + bp - Ts$ with $s = c_p \ln T - (c_p - c_v)\ln(p + p_\infty) + q'$, where $b = p_\infty = 0$ for the vapor and $R_v = c_{p,v} - c_{v,v}$. Dividing (164) by T gives the explicit form

$$R_v \ln p_{vsat} = c_{p,l} - c_{p,v} + q'_v - q'_l + \frac{q_l - q_v + b\,p_l}{T} + (c_{p,v} - c_{p,l})\ln T + (c_{p,l} - c_{v,l})\ln(p_l + p_\infty), \tag{166}$$

in which the liquid pressure enters through b and p_∞. The seed is vapor at $(p_{vsat}(p_l, T_l), T_l)$, in thermal and chemical equilibrium with the liquid, so by (163) $a = (p_{vsat} - p_l + p_j)\,v_v/T_l > 0$; the mechanical step then relaxes the pressure difference. The seed occupies its own volume $\delta m\, v_v$, taken from the liquid: $\delta V_l = -\delta m\, v_v$, $\delta E_l = -\delta m\,(e_v - p_j v_v)$, $\delta m_l = -\delta m$, and the gas receives $\delta m\,[v_v, e_v, 1]^T$.

Condensation. The parent is the gas and the seed is liquid. Define the saturation temperature $T_{sat}(p_l, p_v)$ by

$$g_l(p_l, T_{sat}) = g_v(p_v, T_{sat}), \tag{167}$$

the temperature at which liquid at p_l is in chemical equilibrium with the vapor at the partial pressure p_v. At fixed p_l and p_v, by (4), $\partial(g_v - g_l)/\partial T = s_l - \bar{s}_v$, and at T_{sat}, where $g_l = g_v$, $\bar{s}_v - s_l = (\bar{h}_v - h_l)/T_{sat}$ with $\bar{h}_v = h_v(T)$ of the ideal-gas vapor. On the branch where the latent heat $\bar{h}_v - h_l$ is positive, $g_v > g_l$ if and only if $T < T_{sat}$, and (162), with the liquid at $p_g + p_j$, is

$$T_g < T_{sat}(p_g + p_j, p_v). \tag{168}$$

For the NASG liquid, (167) is (166) with $p_{vsat} = p_v$, solved for T by Newton's method on the branch where its right-hand side increases with T; differentiating (164) at fixed p_l gives $d\ln p_{vsat}/dT = (\bar{h}_v - h_l)/(R_v T^2)$, so this is the branch of positive latent heat. The seed is liquid at $(p_g + p_j, T_g)$, in Laplace and thermal equilibrium with the gas, which by the best seed gives the largest $a = [g_v(p_v, T_g) - g_l(p_g + p_j, T_g)]/T_g > 0$. It occupies its own volume $\delta m\, v_l$, taken from the gas: $\delta V_g = -\delta m\, v_l$, $\delta E_g = -\delta m\,(e_l + p_j v_l)$, $\delta m_v = -\delta m$, and the liquid receives $\delta m\,[v_l, e_l, 1]^T$.

Thresholds. (165) and (168) are the thermodynamic limits of metastability. Two model parameters delay nucleation beyond them: cavitation requires in addition $p_l < p_{cav}$, and condensation $T_g < T_{sat} - \Delta T_c$. The defaults $p_{cav} = \infty$ and $\Delta T_c = 0$ nucleate at the thermodynamic limits.

Size of the seed. The state of the seed is held while its mass grows, so the seed has the entropy $\delta m\, s_s$ at every size, and $dS/d\delta m = a(\delta m)$, with a of (161) evaluated for the parent after it has given δm. By the trapezoidal rule, to second order,

$$\delta S = \tfrac{1}{2}\big[a(0) + a(\delta m)\big]\, \delta m, \tag{169}$$

and $a(\delta m) \le a(0)$, since $da/d\delta m = -\boldsymbol{\sigma}^T \boldsymbol{K}_P \boldsymbol{\sigma} \le 0$ by (31). The seed increases S if $a(\delta m) > -a(0)$, and it keeps at least half of the first-order gain $a(0)\, \delta m$ if $a(\delta m) \ge 0$, i.e. if it does not carry the parent past the state at which the seed is in equilibrium with it.

Seed limit. Write $A \equiv T_P a = g_P - B_s$ by (161); since $T_P > 0$, $a(\delta m) \ge 0$ if and only if $A(\delta m) \ge 0$. With the seed state held and p_j constant, $dB_s = v_s\, dp_P - s_s\, dT_P$. Let $\boldsymbol{z}_P = [p_P, T_P, g_P]^T$ and $\boldsymbol{Z}_P \equiv \partial \boldsymbol{z}_P / \partial \boldsymbol{x}_P = -\boldsymbol{G}_P \boldsymbol{K}_P$ by (70) and (31), with rows $\boldsymbol{Z}_{P,1}$, $\boldsymbol{Z}_{P,2}$, $\boldsymbol{Z}_{P,3}$. The parent changes by $d\boldsymbol{x}_P = -\boldsymbol{A}_P \boldsymbol{\sigma}\, d\delta m$, so

$$\frac{dA}{d\delta m} = -\big(\boldsymbol{Z}_{P,3} - v_s \boldsymbol{Z}_{P,1} + s_s \boldsymbol{Z}_{P,2}\big) \cdot \boldsymbol{A}_P \boldsymbol{\sigma}. \tag{170}$$

The columns of $\boldsymbol{Z}_P$ for V_P and E_P are (108). The mass column is (109) and (110) for the gas; for the liquid it follows from $\boldsymbol{Z}_l \boldsymbol{x}_l = 0$, since scaling the extensive state does not change the intensive one. Linearizing A, the seed is limited by

$$\delta m \le \frac{A(0)}{|dA/d\delta m|} \quad \text{when } \frac{dA}{d\delta m} < 0, \tag{171}$$

evaluated at the parent state before the seed. Then $a(\delta m) = 0$ to first order and $\delta S = \frac{1}{2}\, a(0)\, \delta m$ to second order by (169): the seed does not carry the parent past the state at which it is in equilibrium with the seed.

Cavitation. For the liquid parent, $dg_l = v_l\, dp_l - s_l\, dT_l$ by (4), so $\boldsymbol{Z}_{l,3} = v_l \boldsymbol{Z}_{l,1} - s_l \boldsymbol{Z}_{l,2}$, and (170) is $dA/d\delta m = -(v_v - v_l)\, dp_l/d\delta m + (s_v - s_l)\, dT_l/d\delta m$. The liquid gives the volume $\delta m\, v_v$ and the mass δm, so at fixed mass it is compressed by $\delta m\, (v_v - v_l)$ and gives the energy $\delta m\, (e_v - p_j v_v - e_l)$; by (108) its pressure rises by $dp_l/d\delta m = [K_{s,l}(v_v - v_l) - \Gamma_l(e_v - p_j v_v - e_l + p_l(v_v - v_l))]/V_l$. With $A(0) = (p_{vsat} - p_l + p_j)\, v_v$ by (163), neglecting the temperature and Γ_l terms and v_l against v_v, which holds for a stiff liquid, (171) becomes

$$\delta m \le \frac{\rho_v V_l\, (p_{vsat} - p_l + p_j)}{K_{s,l}}\,: \tag{172}$$

the liquid may be compressed at most to $p_{vsat} + p_j$, where the seed would be in Laplace equilibrium with it.

Condensation. For the gas parent, (170) collects two effects. The gas loses vapor, which lowers g_v by $\mathcal{K}_{CC}\, \delta m$ by (110), and it is heated at nearly fixed volume by the latent heat that the seed leaves in it. At fixed V_g and masses, (108) and (114) give $dT_g = dE_g/C_{v,g}$, $dp_g = \Gamma_g\, dE_g/V_g$ and $dg_v = (R_v - \bar{s}_v)\, dE_g/C_{v,g}$, so the heating changes A by $[(R_v - \bar{s}_v + s_l)/C_{v,g} - v_l \Gamma_g/V_g]\, dE_g$, which is negative when $T(\bar{s}_v - s_l)$, the latent heat, exceeds $R_v T$. Both effects decrease A, and (171) stops the seed before they cancel $A(0)$.

Limits. Both seeds are further limited to a volume fraction ϵ_n of the cell, $\delta m \le \epsilon_n \rho_s$ with $\epsilon_n = 0.01$, and to half of the mass that the parent gives: $\delta m \le m_l/2$ for cavitation and $\delta m \le \min(m_v, m_g)/2$ for condensation.

Relaxation from the seed. Nucleation is tested only when the chemical step is activated, in cells with $\alpha_l < \epsilon_n$ or $\alpha_l > 1 - \epsilon_n$: cavitation if $\alpha_l > 1/2$, condensation otherwise. A dry gas cannot condense, since $g_v \to -\infty$ as $m_v \to 0$. In a near-pure cell the seed is added to the traces of its own phase, and the merged phase starts the relaxation. The seed conserves exactly

$$\delta\alpha_l + \delta\alpha_g = 0, \qquad \delta m_l + \delta m_g = 0, \qquad \delta m_a = 0, \qquad \delta E_l + \delta E_g + p_j\,\delta\alpha_l = 0, \tag{173}$$

as the steps in (158); the interface of the seed itself is not resolved and carries no energy. The steps M, T, C then run from the seeded state. With instantaneous steps the end state is the unique $\boldsymbol{F} = 0$ of the definiteness of the stiffness matrices and does not depend on the seed, which only sets the start of the iteration of the stages. If the stages do not reach it, the seed is undone and the cell is returned unchanged. With a finite chemical rate the seed is a mass transfer not limited by the rate law (140), and the result of the step depends on δm; at $\nu = 0$ no seed is created.

Relation to Pelanti's models. Pelanti's methods [1, 3] do not create a phase: the absent phase is given a small volume fraction everywhere in the initial condition, an initial vapor content in the liquid being "numerically necessary to model cavitation" [3, Sec. 5.1], and the mass transfer is activated where $T_l > T_{sat}(p)$ or where $\min(\alpha_l, \alpha_v)$ exceeds a tolerance [3, Sec. 5]. Her vapor is a separate phase at the mixture pressure p, so $T_l > T_{sat}(p)$ is (165) at $p_l = p$ and $p_j = 0$. The seed replaces this initial content of the absent phase. The opposite situation, a phase that vanishes during the transfer, she treats by setting its volume fraction to a tolerance ϵ_α and rescaling the rate [3, eq. (44)].

Open: complete evaporation. When the root $\boldsymbol{F} = 0$ lies beyond the available liquid mass, the maximum of S is on the boundary $m_l = 0$ and the chemical stage leaves the admissible states; the framework does not yet treat this end state.

1.1.7 Derivation in the phase variables

The steps of the sequential relaxation can also be derived in the phase variables, without the transfer coordinates, as in Pelanti [1]. The derivation below gives the same stiffnesses and increments as (115)–(127).

In mechanical relaxation step, mass of phases remain unchanged and the only energy exchange is due to the volume change

$$dm_k = 0, \quad dE_k = -p_{I,k}\,dV_k, \qquad p_{I,l} = p_I, \quad p_{I,g} = p_I - p_j, \tag{174}$$

where the gas works against $p_I - p_j$ so that $E_l + E_g + \sigma A$ is conserved. This is the direction $\boldsymbol{d}_M$. Inserting the above conditions into equation (14),

$$dS_k = \frac{(p_k - p_{I,k})}{T_k}\,dV_k. \tag{175}$$

Mechanical equilibrium as a constrained entropy maximum. Summing (175) over the phases with $dV_g = -dV_l$,

$$dS = dS_l + dS_g = \left[\frac{p_l - p_I}{T_l} + \frac{p_I - p_j - p_g}{T_g}\right] dV_l = \tilde{F}_M\,dV_l, \tag{176}$$

with $\tilde{F}_M$ of (65). In this step V_l is the only free variable: E_l follows from the work and m_l is fixed. Along the path $d\tilde{V}_l = dV_l$, and S is therefore maximal where $(\partial S/\partial \tilde{V}_l)_{\tilde{E}_l, \tilde{m}_l} = \tilde{F}_M = 0$.

Each term of (176) is the entropy generated by one phase working against the interface. The liquid term is positive when the liquid expands ($dV_l > 0$) against a lower interface pressure ($p_I < p_l$); the gas term is positive when the gas is compressed ($dV_g < 0$) by a higher one ($p_I - p_j > p_g$). The temperatures only scale how much entropy each phase generates; the sign of each term is set by its own pressure difference. To see that the temperatures drop out of the condition $\tilde{F}_M = 0$, insert the interface pressure written as in (78), $p_l - p_I = w_M \hat{F}_M$ and $p_I - p_g - p_j = (1 - w_M)\hat{F}_M$ with $w_M \in [0, 1]$:

$$\tilde{F}_M = \frac{w_M \hat{F}_M}{T_l} + \frac{(1 - w_M)\hat{F}_M}{T_g} = \hat{F}_M \Big(\frac{w_M}{T_l} + \frac{1 - w_M}{T_g} \Big) = \frac{\hat{F}_M}{T_M}. \tag{177}$$

Since w_M and $1 - w_M$ are non-negative and sum to one, and $T_l, T_g > 0$, $1/T_M$ lies between $1/T_g$ and $1/T_l$ and is positive. Along the mechanical path $(\partial S/\partial \tilde{V}_l)_{\tilde{E}_l, \tilde{m}_l} = \hat{F}_M/T_M$: S increases while the liquid expands at $\hat{F}_M > 0$ or contracts at $\hat{F}_M < 0$, and is stationary only at

$$\hat{F}_M = p_l - p_g - p_j = 0. \tag{178}$$

The entropy maximum of mechanical relaxation is therefore the Laplace condition $p_l - p_g = p_j$, for any T_l and T_g; the temperatures enter only the positive factor $1/T_M$. This uses that p_I lies between $p_g + p_j$ and p_l [1, eq. (3)]. If p_I were instead a fixed number, $\tilde{F}_M = 0$ would give $(p_l - p_I)/T_l = (p_g + p_j - p_I)/T_g$, a state where p_I is above both p_l and $p_g + p_j$ or below both, which violates that condition unless $p_l = p_I = p_g + p_j$.

The mechanical end state is not $\boldsymbol{F} = 0$: the condition (25) holds only when V_l, E_l and m_l can all change independently, i.e. at full equilibrium. In particular, the first component of $\boldsymbol{F}$, $F_1 = p_l/T_l - (p_g + p_j)/T_g$, is the force of a volume exchange at fixed E_l, along $[1, 0, 0]^T = \boldsymbol{d}_M - p_I \boldsymbol{d}_T$, which needs the heat $p_I\, dV_l$ received by the liquid ($d\tilde{E}_l = -p_I\, dV_l$) alongside the work: equal p/T is the mechanical condition only when heat is exchanged as well. At the mechanical end state $F_1 = p_I(1/T_l - 1/T_g) \neq 0$ is balanced by the energy component, $\tilde{F}_M = F_1 - p_I F_2 = 0$. Pressure equilibrium is one condition for two unknowns, V_l and the split of the energy, so the mechanical end state depends on p_I along the path; the thermal and chemical end states have as many conditions as unknowns.

Pressure as a function of entropy and volume is

$$dp_k = \left(\frac{\partial p_k}{\partial V_k} \right)_{S_k} dV_k + \left(\frac{\partial p_k}{\partial S_k} \right)_{V_k} dS_k, \tag{179}$$

which using equation (175), can be rearranged to

$$dp_k = \left[\left(\frac{\partial p_k}{\partial V_k} \right)_{S_k} + \left(\frac{\partial p_k}{\partial S_k} \right)_{V_k} \left(\frac{p_k - p_{I,k}}{T_k} \right) \right] dV_k. \tag{180}$$

The isentropic bulk modulus is defined as

$$K_s = \rho \left(\frac{\partial p}{\partial \rho} \right)_s = -v \left(\frac{\partial p}{\partial v} \right)_s. \tag{181}$$

At constant mass we have

$$\left(\frac{\partial p_k}{\partial V_k} \right)_{S_k} = \frac{-K_s}{V} \tag{182}$$

The Grüneisen coefficient is defined as

$$\Gamma = v \left(\frac{\partial p}{\partial e} \right)_v. \tag{183}$$

Using equation (1) and by chain rule we obtain

$$\left(\frac{\partial p}{\partial s}\right)_v = \left(\frac{\partial p}{\partial e}\right)_v \left(\frac{\partial e}{\partial s}\right)_v = \frac{\Gamma}{v}\, T, \tag{184}$$

which leads to the following relation

$$\left(\frac{\partial p}{\partial S}\right)_V = \frac{\Gamma}{V}\, T. \tag{185}$$

Plugging the obtained partial derivatives of pressure back into equation (180) we get

$$dp_k = \left[-\frac{K_{s,k}}{V_k} + \frac{\Gamma_k}{V_k}\,(p_k - p_{I,k})\right] dV_k, \tag{186}$$

which relates phasic pressure and phasic volume change for mechanical relaxation step. Compliance is defined as $W_k = V/K_s$. With this definition, the above equation can be turned into

$$dp_k = -\frac{dV_k}{W_k^{eff}}, \tag{187}$$

where

$$W_k^{eff} = \frac{W_k}{1 - \Gamma_k\,(p_k - p_{I,k})\,/K_{s,k}}. \tag{188}$$

Mechanical relaxation requires that

$$\left(p_l^0 + \delta p_l^M\right) - \left(p_g^0 + \delta p_g^M\right) = p_j. \tag{189}$$

Therefore,

$$\hat{F}_M^0 + \delta p_l^M - \delta p_g^M = 0, \tag{190}$$

where

$$\hat{F}_M^0 = p_l^0 - p_g^0 - p_j \tag{191}$$

is the mechanical component of $\hat{\boldsymbol{F}}$ at the start state. Using equations (186), (191), and volume conservation, solution to mechanical relaxation step is found

$$p_l^* = p_l^0 + \delta p_l^M, \qquad p_g^* = p_g^0 + \delta p_g^M, \tag{192}$$

$$\delta p_l^M = -\phi_l^M\, \hat{F}_M^0, \qquad \delta p_g^M = +\phi_g^M\, \hat{F}_M^0, \tag{193}$$

$$\phi_l^M \equiv \frac{W_g^{eff}}{W_l^{eff} + W_g^{eff}}, \qquad \phi_g^M \equiv \frac{W_l^{eff}}{W_l^{eff} + W_g^{eff}}. \tag{194}$$

In terms of the transferred volume $\delta\tilde{V}_l = \delta V_l = -\delta V_g$, (187) gives $d\hat{F}_M = dp_l - dp_g = -(1/W_l^{eff} + 1/W_g^{eff})\, d\tilde{V}_l$, so by (98) and (101), with W_k^{eff} at the start state 0,

$$\mathcal{K}^M = \frac{1}{W_l^{eff}} + \frac{1}{W_g^{eff}}, \qquad \delta\tilde{V}_l = \frac{\hat{F}_M^0}{\mathcal{K}^M}, \qquad \delta E_l = -p_{I,l}\,\delta\tilde{V}_l, \qquad \delta E_g = p_{I,g}\,\delta\tilde{V}_l, \tag{195}$$

This is (115), with $1/W_k^{eff}$ its phase terms, and it gives $\delta p_l^M = -\delta\tilde{V}_l/W_l^{eff} = -\phi_l^M \hat{F}_M^0$. $1/W_k^{eff} = [\Gamma_k(p_{I,k} - p_k) + K_{s,k}]/V_k$ is Pelanti's $1/\xi_k^p$ [1, eq. (51)], where $p_j = 0$. The corresponding temperature change would be

$$T_l^* = T_l^0 + \delta T_l^M, \qquad T_g^* = T_g^0 + \delta T_g^M, \tag{196}$$

$$\delta T_k^M = -W_k^{eff}\left[\frac{p_k - p_{I,k}}{C_{v,k}} - \frac{\Gamma_k T_k}{V_k}\right]\delta p_k^M. \tag{197}$$

The derivation of δT_k is as follows. For each phase at frozen composition we have

$$dT = \left(\frac{\partial T}{\partial V}\right)_S dV + \left(\frac{\partial T}{\partial S}\right)_V dS. \tag{198}$$

From equation (1) we get

$$c_v = \left(\frac{\partial e}{\partial T}\right)_v = T\left(\frac{\partial s}{\partial T}\right)_v. \tag{199}$$

Therefore we have

$$\left(\frac{\partial T}{\partial S}\right)_V = \frac{T}{C_v}, \quad C_v = mc_v. \tag{200}$$

From equation (1) we get

$$T = \left(\frac{\partial e}{\partial s}\right)_v, \quad p = -\left(\frac{\partial e}{\partial v}\right)_s, \tag{201}$$

and

$$\left(\frac{\partial T}{\partial v}\right)_s = -\left(\frac{\partial p}{\partial s}\right)_v = -\frac{\Gamma T}{v}, \tag{202}$$

where equation (184) is used. The temperature differential can then be written as

$$dT = -\frac{\Gamma T}{V}dV + \frac{T}{C_v}dS, \tag{203}$$

which by using equation (175) can be rearranged to

$$dT_k = \left(\frac{p_k - p_{I,k}}{C_{v,k}} - \frac{\Gamma_k T_k}{V_k}\right)dV_k, \tag{204}$$

In thermal relaxation step, phasic masses remain unchanged while heat is exchanged between the phases. Phasic pressure are allowed to change such that at the end of the relaxation, Thermal and mechanical equilibria are achieved. The thermal component of $\hat{\boldsymbol{F}}$ at the start state is

$$\hat{F}_T^* = T_l^* - T_g^*. \tag{205}$$

Thermal equilibrium as a constrained entropy maximum. In this step the masses are fixed and the pressure equilibrium $p_l = p_g + p_j$ of the mechanical step is held. The free variable is the heat $\delta\tilde{E}_l$ given by the liquid; the volumes adjust to hold the pressure equilibrium. The liquid gives the heat and works at its own pressure, and the energy constraint $dE_g = -dE_l - p_j\,dV_l$ then gives the gas energy:

$$dE_l = -d\tilde{E}_l - p_l\,dV_l, \qquad dE_g = d\tilde{E}_l + (p_l - p_j)\,dV_l = d\tilde{E}_l - p_g\,dV_g. \tag{206}$$

This is the direction $\boldsymbol{d}_T$ plus the work along $\boldsymbol{d}_M$ with $p_I = p_l$, the only value between $p_g + p_j$ and p_l at pressure equilibrium. With fixed masses, (14) gives $dS_k = (dE_k + p_k\,dV_k)/T_k$ for each phase. Substituting (206),

$$dS_l = \frac{-d\tilde{E}_l - p_l\,dV_l + p_l\,dV_l}{T_l} = -\frac{d\tilde{E}_l}{T_l}, \qquad dS_g = \frac{d\tilde{E}_l - p_g\,dV_g + p_g\,dV_g}{T_g} = \frac{d\tilde{E}_l}{T_g}. \tag{207}$$

The work terms cancel because each phase works at its own pressure: at pressure equilibrium the volume exchange generates no entropy, and only the heat does. Summing,

$$dS = dS_l + dS_g = \Big(\frac{1}{T_g} - \frac{1}{T_l}\Big) d\tilde{E}_l = \tilde{F}_T \, d\tilde{E}_l, \tag{208}$$

with $\tilde{F}_T$ of (65). Over the common denominator,

$$\tilde{F}_T = \frac{T_l - T_g}{T_l T_g} = \frac{\hat{F}_T}{T_l T_g}. \tag{209}$$

$T_l, T_g > 0$, so $1/(T_l T_g)$ is positive and $dS = \hat{F}_T \, d\tilde{E}_l/(T_l T_g)$: S increases while heat flows from the hotter phase to the colder one ($d\tilde{E}_l > 0$ at $\hat{F}_T > 0$, $d\tilde{E}_l < 0$ at $\hat{F}_T < 0$), and is stationary only at

$$\hat{F}_T = T_l - T_g = 0. \tag{210}$$

The entropy maximum of thermal relaxation is therefore the temperature equilibrium; the temperatures enter only the positive factor $1/(T_l T_g)$. Unlike (177), this factorization needs no interface closure. With the pressure equilibrium held, the end state of the thermal step is the maximum of S over all states with fixed masses: $p_l = p_g + p_j$ and $T_l = T_g$, two conditions for the two unknowns V_l and E_l.

Thermo-mechanical equilibrium requires that

$$p_l^{**} - p_g^{**} = p_j, \quad T_l^{**} = T_g^{**} \tag{211}$$

Note that since mechanical relaxation leaves pressure equilibrium through $p_l^* - p_g^* = p_j$, pressure change during thermal relaxation step is identical for liquid and gas phases

$$\delta p_l^T = \delta p_g^T = \delta p^T. \tag{212}$$

Temperature change reads

$$T_l^{**} = T_l^* + \delta T_l^T, \quad T_g^{**} = T_g^* + \delta T_g^T, \quad \hat{F}_T^* = \delta T_g^T - \delta T_l^T. \tag{213}$$

Conservation requires that

$$dV_l + dV_g = 0 \tag{214}$$

and

$$d(E_l + E_g + \sigma A) = 0 \tag{215}$$

$$dH_l + dH_g - (p_l - p_g - p_j) dV_l - (V_l + V_g) dp = 0 \tag{216}$$

$$dH_l + dH_g - V dp = 0 \tag{217}$$

The differential form of volume can be written as

$$dV = \left(\frac{\partial V}{\partial T}\right)_p dT + \left(\frac{\partial V}{\partial p}\right)_T dp, \tag{218}$$

by using thermal expansion coefficient

$$\alpha_p = \frac{1}{v}\left(\frac{\partial v}{\partial T}\right)_p, \tag{219}$$

isothermal bulk modulus

$$K_T = \rho \left(\frac{\partial p}{\partial \rho}\right)_T = -v \left(\frac{\partial p}{\partial v}\right)_T, \tag{220}$$

and phasic mass conservation can be reduced to the following form

$$dV = V\alpha_p dT - \frac{V}{K_T} dp. \tag{221}$$

At constant pressure, equation (3) gives $dh = Tds$. On the other hand,

$$ds = \left(\frac{\partial s}{\partial T}\right)_p dT + \left(\frac{\partial s}{\partial p}\right)_T dp. \tag{222}$$

Combining the above relation with $dh = Tds$ at constant pressure we get

$$\left(\frac{\partial h}{\partial T}\right)_p = T \left(\frac{\partial s}{\partial T}\right)_p \Rightarrow \left(\frac{\partial s}{\partial T}\right)_p = \frac{c_p}{T} \tag{223}$$

Equation (4) gives

$$v = \left(\frac{\partial g}{\partial p}\right)_T \quad and \quad s = -\left(\frac{\partial g}{\partial T}\right)_p. \tag{224}$$

As a result,

$$\left(\frac{\partial s}{\partial p}\right)_T = \frac{\partial}{\partial p}\left[-\left(\frac{\partial g}{\partial T}\right)_p\right]_T = -\frac{\partial}{\partial T}\left[\left(\frac{\partial g}{\partial p}\right)_T\right]_p = -\left(\frac{\partial v}{\partial T}\right)_p = -v\alpha_p. \tag{225}$$

Therefore, the differential form of entropy would read

$$ds = \frac{c_p}{T} dT - v\alpha_p dp. \tag{226}$$

Substituting the above equation into equation (3)

$$dh = T\left(\frac{c_p}{T} dT - v\alpha_p dp\right) + v dp = c_p dT + v\left(1 - T\alpha_p\right) dp, \tag{227}$$

which can be transferred into extensive form given phasic mass conservation through thermal relaxation

$$dH = C_p dT + V\left(1 - T\alpha_p\right) dp, \tag{228}$$

where $C_p = mc_p$. Using equation (218) and that the total volume does not change in thermal relaxation

$$\left(V_l \alpha_{p,l}\, dT_l - \frac{V_l}{K_{T,l}} dp\right) + \left(V_g \alpha_{p,g}\, dT_g - \frac{V_g}{K_{T,g}} dp\right) = 0, \tag{229}$$

which can be rearranged into

$$V_l \alpha_{p,l}\, dT_l + V_g \alpha_{p,g}\, dT_g - W_T dp = 0, \tag{230}$$

where

$$W_T \equiv \frac{V_l}{K_{T,l}} + \frac{V_g}{K_{T,g}}. \tag{231}$$

Now substitute equation (228) into energy conservation

$$C_{p,l}dT_l + V_l\,(1 - T_l\alpha_{p,l})\,dp + C_{p,g}dT_g + V_g\,(1 - T_g\alpha_{p,g})\,dp - (V_l + V_g)dp = 0, \tag{232}$$

which is simplified to

$$C_{p,l}dT_l + C_{p,g}dT_g - \mathcal{A}_{pT}dp = 0, \tag{233}$$

where

$$\mathcal{A}_{pT} \equiv V_l T_l \alpha_{p,l} + V_g T_g \alpha_{p,g}. \tag{234}$$

Applied to the increments δT_l^T, δT_g^T, δp^T of the step, with the coefficients at the start state $*$, equations (213), (230), and (233) form a system of 3 equations and 3 unknowns, whose solution is

$$T_l^{**} = T_l^* + \delta T_l^T, \qquad T_g^{**} = T_g^* + \delta T_g^T, \tag{235}$$

$$\delta T_l^T = -\phi_l^T \hat{F}_T^*, \quad \delta T_g^T = +\phi_g^T \hat{F}_T^* \tag{236}$$

$$\phi_l^T = \frac{W_T C_{p,g} - \mathcal{A}_{pT} V_g \alpha_{p,g}}{\mathcal{D}}, \quad \phi_g^T = \frac{W_T C_{p,l} - \mathcal{A}_{pT} V_l \alpha_{p,l}}{\mathcal{D}}, \tag{237}$$

and the corresponding pressure would be

$$p_l^{**} = p_l^* + \delta p^T, \quad p_g^{**} = p_g^* + \delta p^T, \quad \delta p^T = \frac{V_l \alpha_{p,l} \delta T_l^T + V_g \alpha_{p,g} \delta T_g^T}{W_T}, \tag{238}$$

where the coefficients used in the above equations are listed below

$$\mathcal{A}_p \equiv V_l \alpha_{p,l} + V_g \alpha_{p,g}, \quad C_p \equiv C_{p,l} + C_{p,g}, \quad \mathcal{D} \equiv C_p W_T - \mathcal{A}_p \mathcal{A}_{pT}. \tag{239}$$

In terms of the heat $\delta\tilde{E}_l$ of (206), with $dH_k = dE_k + p_k\,dV_k + V_k\,dp$ and (228),

$$-d\tilde{E}_l = C_{p,l}\,dT_l - a_l\,dp, \qquad d\tilde{E}_l = C_{p,g}\,dT_g - a_g\,dp, \qquad a_k \equiv V_k T_k \alpha_{p,k}. \tag{240}$$

Their sum is (233), since $\mathcal{A}_{pT} = a_l + a_g$. Write (230) as $b_l\,dT_l + b_g\,dT_g = W_T\,dp$ with $b_k \equiv V_k \alpha_{p,k}$, so that $\mathcal{A}_p = b_l + b_g$. Substituting dT_l and dT_g from (240),

$$\left(\frac{\partial p}{\partial \tilde{E}_l}\right)_{\hat{F}_M, \tilde{m}_l} = -\frac{1}{W_E}\Big(\frac{b_l}{C_{p,l}} - \frac{b_g}{C_{p,g}}\Big), \qquad W_E \equiv W_T - \frac{a_l b_l}{C_{p,l}} - \frac{a_g b_g}{C_{p,g}}, \tag{241}$$

$$\left(\frac{\partial T_l}{\partial \tilde{E}_l}\right)_{\hat{F}_M, \tilde{m}_l} = \frac{1}{C_{p,l}}\Big[-1 + a_l \left(\frac{\partial p}{\partial \tilde{E}_l}\right)_{\hat{F}_M, \tilde{m}_l}\Big], \qquad \left(\frac{\partial T_g}{\partial \tilde{E}_l}\right)_{\hat{F}_M, \tilde{m}_l} = \frac{1}{C_{p,g}}\Big[1 + a_g \left(\frac{\partial p}{\partial \tilde{E}_l}\right)_{\hat{F}_M, \tilde{m}_l}\Big], \tag{242}$$

and, by (98) and (101),

$$\begin{aligned} \mathcal{K}^T &= -\left(\frac{\partial \hat{F}_T}{\partial \tilde{E}_l}\right)_{\hat{F}_M, \tilde{m}_l} = \left(\frac{\partial T_g}{\partial \tilde{E}_l}\right)_{\hat{F}_M, \tilde{m}_l} - \left(\frac{\partial T_l}{\partial \tilde{E}_l}\right)_{\hat{F}_M, \tilde{m}_l} \\ &= \frac{1}{C_{p,l}} + \frac{1}{C_{p,g}} + \frac{1}{W_E}\Big(\frac{a_l}{C_{p,l}} - \frac{a_g}{C_{p,g}}\Big)\Big(\frac{b_l}{C_{p,l}} - \frac{b_g}{C_{p,g}}\Big), \qquad \delta\tilde{E}_l = \frac{\hat{F}_T^*}{\mathcal{K}^T}. \end{aligned} \tag{243}$$

This step reproduces the thermal solution above:

$$\phi_l^T = -\frac{1}{\mathcal{K}^T}\left(\frac{\partial T_l}{\partial \tilde{E}_l}\right)_{\hat{F}_M, \tilde{m}_l}, \qquad \phi_g^T = \frac{1}{\mathcal{K}^T}\left(\frac{\partial T_g}{\partial \tilde{E}_l}\right)_{\hat{F}_M, \tilde{m}_l}, \qquad \delta p^T = \left(\frac{\partial p}{\partial \tilde{E}_l}\right)_{\hat{F}_M, \tilde{m}_l} \delta\tilde{E}_l. \tag{244}$$

These are the increments (121) of the thermal step, and (243) equals (119).

In chemical relaxation step, liquid and gas exchange mass while forcing thermo-mechanical equilibrium and as a result, liquid and gas phases share the same δT and δp.

$$p_l^{\otimes} = p_l^{**} + \delta p^C, \quad p_g^{\otimes} = p_g^{**} + \delta p^C, \quad p_l^{\otimes} - p_g^{\otimes} = p_j \tag{245}$$

$$T_l^{\otimes} = T_l^{**} + \delta T^C, \quad T_g^{\otimes} = T_g^{**} + \delta T^C, \quad T_l^{\otimes} = T_g^{\otimes}. \tag{246}$$

The chemical component of $\hat{\boldsymbol{F}}$ at the start state and the chemical potential equilibrium are

$$\hat{F}_C^{**} = g_l^{**} - g_v^{**}, \quad g_l^{\otimes} = g_v^{\otimes}. \tag{247}$$

Chemical equilibrium as a constrained entropy maximum. In this step the pressure and temperature equilibria $p_l = p_g + p_j$ and $T_l = T_g = T$ are held and mass crosses the interface. The free variable is the mass $\delta\tilde{m}_l = -\delta m_l$ given by the liquid to the gas, with $dm_v = d\tilde{m}_l$; V_l and E_l adjust to hold the two equilibria. Whatever these adjustments are, (24) with the components (57) gives

$$dS = F_1\, dV_l + F_2\, dE_l - F_3\, d\tilde{m}_l. \tag{248}$$

At the held equilibria the first two components vanish,

$$F_1 = \frac{p_l}{T} - \frac{p_g + p_j}{T} = \frac{p_l - p_g - p_j}{T} = 0, \qquad F_2 = \frac{1}{T} - \frac{1}{T} = 0, \qquad F_3 = -\frac{g_l - g_v}{T}, \tag{249}$$

so the volume and energy that accompany the mass generate no entropy, and

$$dS = \frac{g_l - g_v}{T}\, d\tilde{m}_l = \tilde{F}_C\, d\tilde{m}_l, \qquad \tilde{F}_C = \frac{\hat{F}_C}{T}. \tag{250}$$

The same follows from $\tilde{F}_C$ of (65): with $p_g + p_j = p_l$ and $T_g = T_l = T$, and by (2) $h_l = e_l + p_l v_l$ and $g_l = h_l - T s_l$,

$$\tilde{F}_C = \frac{e_l + p_l v_l - g_v}{T} - s_l = \frac{h_l - T s_l - g_v}{T} = \frac{g_l - g_v}{T} = \frac{\hat{F}_C}{T}. \tag{251}$$

$T > 0$, so S increases while mass goes from the phase with the higher chemical potential to the one with the lower ($d\tilde{m}_l > 0$, evaporation, at $\hat{F}_C > 0$; $d\tilde{m}_l < 0$, condensation, at $\hat{F}_C < 0$), and is stationary only at

$$\hat{F}_C = g_l - g_v = 0. \tag{252}$$

The entropy maximum of chemical relaxation is therefore the equality of chemical potentials, with g_v at the vapor partial pressure; the temperature enters only the positive factor $1/T$. Because $F_1 = F_2 = 0$, the result does not depend on the volume and energy that accompany the mass, i.e. on $\boldsymbol{d}_C$. With the held equilibria, the end state of the chemical step is the maximum of S over all exchanges, $\boldsymbol{F} = 0$.

During chemical relaxation, total volume, energy, and mass stay unchanged.

$$dV_l + dV_g = 0, \quad dE_l + dE_g + p_j dV_l = 0, \quad dm_l + dm_v = 0 \tag{253}$$

With mass changing, the differential of volume and enthalpy would be

$$dV = d(mv) = mdv + vdm = V\alpha_p dT - \frac{V}{K_T} dp + vdm \tag{254}$$

$$dH = d(mh) = mdh + hdm = C_p dT + V\left(1 - T\alpha_p\right) dp + hdm \tag{255}$$

For liquid, which is assumed to be single species we have

$$dV_l = V_l \alpha_{p,l} dT_l - \frac{V_l}{K_{T,l}} dp_l + v_l dm_l \tag{256}$$

and

$$dH_l = C_{p,l} dT_l + V_l\left(1 - T_l \alpha_{p,l}\right) dp_l + h_l dm_l. \tag{257}$$

However, for the gas mixture, which has two species the relations would read

$$dV_g = V_g \alpha_{p,g} dT_g - \frac{V_g}{K_{T,g}} dp_g + \left(\frac{\partial V_g}{\partial m_v}\right)_{p_g, T_g, m_a} dm_v \tag{258}$$

and

$$dH_g = C_{p,g} dT_g + V_g\left(1 - T_g \alpha_{p,g}\right) dp_g + \left(\frac{\partial H_g}{\partial m_v}\right)_{p_g, T_g, m_a} dm_v. \tag{259}$$

Plugging the above dV_k and dH_k relations into volume and energy invariance equations, with the increments of the step δp^C, δT^C and the mass given by the liquid $\delta \tilde{m}_l = -\delta m_l = \delta m_v$, we get

$$\mathcal{A}_p\, \delta T^C - W_T\, \delta p^C - \Delta v\, \delta \tilde{m}_l = 0 \tag{260}$$

and

$$C_p\, \delta T^C - \mathcal{A}_{pT}\, \delta p^C - \Delta h\, \delta \tilde{m}_l = 0, \tag{261}$$

where

$$\bar{v}_v = \left(\frac{\partial V_g}{\partial m_v}\right)_{p_g, T_g, m_a}, \quad \Delta v = v_l - \bar{v}_v, \tag{262}$$

$$\bar{h}_v = \left(\frac{\partial H_g}{\partial m_v}\right)_{p_g, T_g, m_a}, \quad and \quad \Delta h = h_l - \bar{h}_v. \tag{263}$$

Using the ideal gas law

$$V_g = \frac{(m_v R_v + m_a R_a) T_g}{p_g} \rightarrow \bar{v}_v = \left(\frac{\partial V_g}{\partial m_v}\right)_{p_g, T_g, m_a} = \frac{R_v T_g}{p_g} = x_v v_v, \tag{264}$$

$$H_g = m_v h_v(T_g) + m_a h_a(T_g) \rightarrow \bar{h}_v = \left(\frac{\partial H_g}{\partial m_v}\right)_{p_g, T_g, m_a} = h_v \tag{265}$$

Now expanding dG_k as functions of dp_k, dT_k, and dm_k

$$dg_l = -s_l dT_l + v_l dp_l \quad and \quad dg_v = v_v dp_v - s_v dT_g, \tag{266}$$

where $\mu_v = \left(\frac{\partial G_g}{\partial m_v}\right)_{p_g, T_g, m_a}$ and $p_v = x_v p_g$. The mol fraction of vapor in gas phase is

$$x_v = \frac{N_v}{N_v + N_a} = \frac{M_a m_v}{M_a m_v + M_v m_a}, \tag{267}$$

where M_i is the molar mass of species i. Therefore

$$dx_v = \frac{M_a M_v m_a}{(M_a m_v + M_v m_a)^2} dm_v = \frac{x_v(1 - x_v)}{m_v} dm_v. \tag{268}$$

Hence,

$$dg_v = v_v x_v dp_g - s_v dT_g + v_v p_g \frac{x_v(1-x_v)}{m_v} dm_v. \tag{269}$$

Finally, the 3×3 system is

$$-W_T\,\delta p^C + \mathcal{A}_p\,\delta T^C - \Delta v\,\delta\tilde{m}_l = 0, \tag{270}$$

$$-\mathcal{A}_{pT}\,\delta p^C + C_p\,\delta T^C - \Delta h\,\delta\tilde{m}_l = 0, \tag{271}$$

$$-\Delta v\,\delta p^C + \Delta s\,\delta T^C + \mathcal{K}_{CC}\,\delta\tilde{m}_l = \hat{F}_C^{**}, \tag{272}$$

where the third row is $\hat{F}_C^{**} + \delta\hat{F}_C = 0$ with $d\hat{F}_C = dg_l - dg_v$, $\Delta s = s_l - \bar{s}_v$, $\bar{s}_v = s_v(T, p_v)$, and

$$\mathcal{K}_{CC} = v_v p_g \frac{x_v(1-x_v)}{m_v} = \frac{R_v T_g (1-x_v)}{m_v}. \tag{273}$$

The solution of the above system is

$$m_l^{\otimes} = m_l^{**} - \delta\tilde{m}_l, \quad m_v^{\otimes} = m_v^{**} + \delta\tilde{m}_l, \tag{274}$$

$$\delta\tilde{m}_l = \phi_l^C \hat{F}_C^{**}, \tag{275}$$

$$\phi_l^C = \frac{1}{\Delta v\Pi - \Delta s\Theta + \mathcal{K}_{CC}}, \tag{276}$$

$$p_l^{\otimes} = p_l^{**} + \delta p^C, \quad p_g^{\otimes} = p_g^{**} + \delta p^C, \tag{277}$$

$$\delta p^C = -\Pi\,\delta\tilde{m}_l, \tag{278}$$

$$T_l^{\otimes} = T_g^{\otimes} = T^{**} + \delta T^C, \quad \delta T^C = -\Theta\,\delta\tilde{m}_l, \tag{279}$$

where Π and Θ of (125), evaluated at the start state $**$, take the closed form

$$\Pi = \frac{C_p\Delta v - \mathcal{A}_p\Delta h}{\mathcal{D}}, \quad \Theta = \frac{\mathcal{A}_{pT}\Delta v - W_T\Delta h}{\mathcal{D}}. \tag{280}$$

The last row of the 3×3 system, with $dp = -\Pi\,d\tilde{m}_l$ and $dT = -\Theta\,d\tilde{m}_l$ along the step, gives $d\hat{F}_C = dg_l - dg_v = -(\Delta v\,\Pi - \Delta s\,\Theta + \mathcal{K}_{CC})\,d\tilde{m}_l$, i.e. $(\partial\hat{F}_C/\partial\tilde{m}_l)_{\hat{F}_M,\hat{F}_T} = -(\Delta v\,\Pi - \Delta s\,\Theta + \mathcal{K}_{CC})$, so by (98) and (101)

$$\mathcal{K}^C = \frac{1}{\phi_l^C} = \Delta v\,\Pi - \Delta s\,\Theta + \mathcal{K}_{CC}, \qquad \delta\tilde{m}_l = \frac{\hat{F}_C^{**}}{\mathcal{K}^C}, \tag{281}$$

which is (127). The volume and energy that accompany $\delta\tilde{m}_l$ follow from the held equilibria, not from $\boldsymbol{d}_C$.

References

[1] M. Pelanti, Arbitrary-rate relaxation techniques for the numerical modeling of compressible two-phase flows with heat and mass transfer, Int. J. Multiphase Flow 153 (2022) 104097.

[2] M. Pelanti, K.-M. Shyue, A mixture-energy-consistent six-equation two-phase numerical model for fluids with interfaces, cavitation and evaporation waves, J. Comput. Phys. 259 (2014) 331–357.

[3] M. Pelanti, Numerical relaxation techniques for mass transfer in three-phase liquid-vapor-gas flows, Comput. Fluids 305 (2026) 106893.